

Quality Testing Analysis and MCDM Techniques to Study the Behaviour of Brain Tumor Drugs Using Distance-Based Topological indices

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Abstract

Brain tumors form when brain cells undergo DNA mutations and grow uncontrollably due to genetic, environmental, or combined factors. This study introduces novel distance-based topological indices derived from the proposed KP-polynomial, targeting drugs used in brain tumor treatment. The main goal of the QSPR (Quantitative Structure-Property Relationship) analysis is to link molecular descriptors with key drug properties like boiling point, flash point, and enthalpy of vaporization. A polynomial regression model assesses the predictive power of these indices, with correlation coefficients indicating the relationship between predicted and experimental values. This theoretical approach aids chemists and healthcare professionals in estimating drug properties without experimental testing and supports drug discovery and advanced decision-making techniques such as CRITIC, TOPSIS, VIKOR, and COPRAS for drug ranking and analysis.

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Brain tumors form when brain cells undergo DNA mutations and grow uncontrollably due to genetic, environmental, or combined factors. This study introduces novel distance-based topological indices derived from the proposed KP-polynomial, targeting drugs used in brain tumor treatment. The main goal of the QSPR (Quantitative Structure-Property Relationship) analysis is to link molecular descriptors with key drug properties like boiling point, flash point, and enthalpy of vaporization. A polynomial regression model assesses the predictive power of these indices, with correlation coefficients indicating the relationship between predicted and experimental values. This theoretical approach aids chemists and healthcare professionals in estimating drug properties without experimental testing and supports drug discovery and advanced decision-making techniques such as CRITIC, TOPSIS, VIKOR, and COPRAS for drug ranking and analysis.

Keywords — Graph polynomial, distance-based, topological indices, Brain tumor treatment drugs.

1 | INTRODUCTION

Brain tumors are among the most complex and debilitating neurological disorders, significantly impacting global health. These tumors arise from abnormal cell proliferation in the brain or its surrounding structures, such as the meninges, cranial nerves, or pituitary gland. They are broadly categorized as primary tumors, originating within the brain, and secondary or metastatic tumors, which result from the spread of cancer from other parts of the body. Primary brain tumors, including gliomas, meningiomas, and medulloblastomas, vary widely in their histological composition, growth rate, and response to treatment. Among these, glioblastoma multiforme (GBM) is the most aggressive and fatal, accounting for a significant proportion of brain tumor-related deaths.

The pathophysiology of brain tumors involves a complex interplay of genetic mutations, epigenetic alterations, and dysregulated cellular signaling pathways. Oncogenes, tumor suppressor genes, and factors such as angiogenesis and immune evasion play pivotal roles in tumor development and progression. Moreover, the unique environment of the central nervous system, characterized by the blood-brain barrier and limited regenerative capacity, poses significant challenges to treatment. Despite advances in neuroimaging and molecular diagnostics, early detection remains difficult, often delaying intervention until the disease has reached an advanced stage[36].

Therapeutic approaches for brain tumors have evolved considerably over the past few decades, encompassing surgical resection, radiotherapy, chemotherapy, and, more recently, targeted therapies and immunotherapy. However, the prognosis for malignant brain tumors remains poor, with survival rates for aggressive forms like GBM averaging only 12 to 18 months despite intensive treatment[37]. Emerging research is focused on uncovering the molecular and genetic underpinnings of brain tumors, paving the way for personalized medicine and innovative therapies such as gene editing, nanotechnology, and oncolytic virotherapy.

The illustration of a brain with a tumor, showcasing the tumor's location and brain structures are given in the Figure1.

Brain tumors are classified into two main categories: primary (originating in the brain) and secondary (metastatic, spreading from other parts of the body). Key types of brain tumors include:

- Gliomas (e.g., glioblastoma, astrocytoma) – tumors arising from glial cells that support neurons.
- Meningiomas – tumors from the meninges, the protective layers around the brain.
- Pituitary tumors – affecting the pituitary gland, influencing hormone production.
- Medulloblastomas – common in children, originating in the cerebellum.
- Schwannomas – arising from Schwann cells, often impacting the auditory nerve.

Diagnostic techniques for brain tumors include:

- MRI (Magnetic Resonance Imaging) – provides detailed images of brain structure and can identify tumor size and location.
- CT scan (Computed Tomography) – useful in emergency settings to detect large tumors or bleeding.
- Biopsy – a sample of tumor tissue is examined to determine type and malignancy.
- PET scan (Positron Emission Tomography) – assesses the metabolic activity of the tumor.
- Neuroimaging with contrast – helps distinguish tumor types based on blood flow patterns.

These methods help guide treatment decisions, ranging from surgery to radiation therapy and chemotherapy. The death rate from brain tumors varies based on several factors, including the type of tumor, its location, and the stage

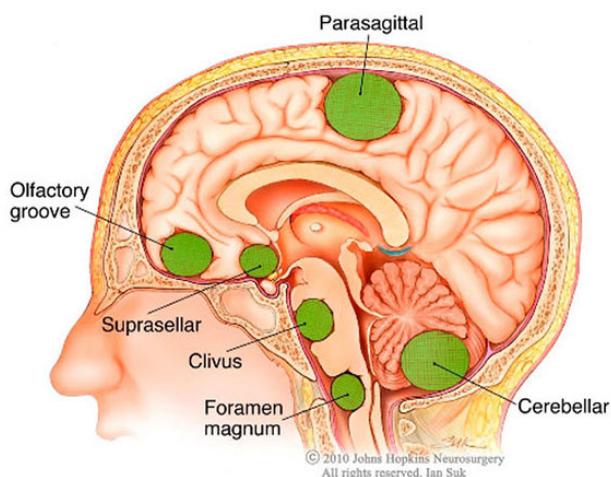


FIGURE 1 Illustration of a brain with a tumor, tumor's location and brain structures.

at which it's diagnosed.

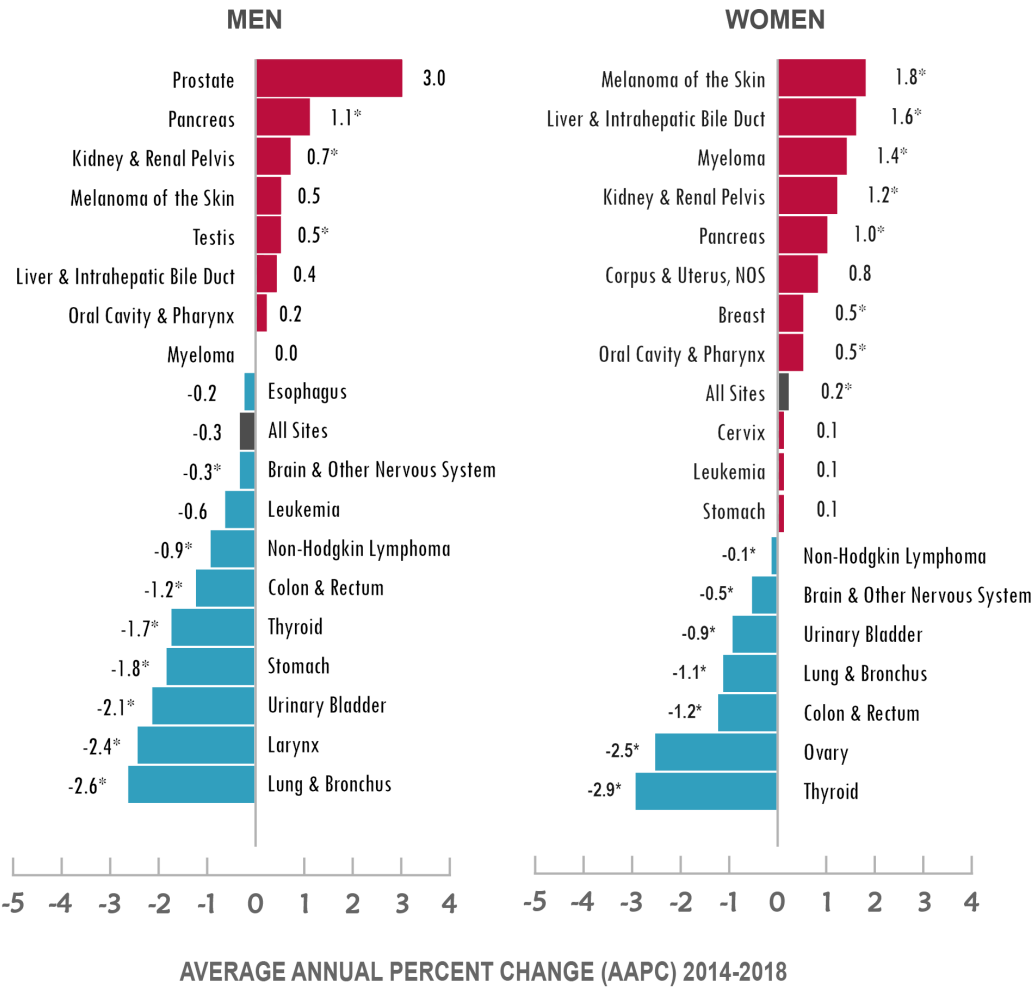
- Glioblastoma – This is one of the most aggressive brain tumors, with a median survival rate of about 12 to 18 months. The 5-year survival rate is typically around 5-10%.
- Meningiomas – These are generally benign and have a much better prognosis, with a 5-year survival rate of about 70-90%. However, malignant forms can be more dangerous.
- Medulloblastomas – While common in children, this tumor type has a 5-year survival rate of about 70%, although it can be lower in older adults or if diagnosed at a later stage.
- Pituitary tumors – Most are benign, and with proper treatment, the prognosis is often good, with a high survival rate. However, if undiagnosed or untreated, they can cause serious complications.

Overall, the prognosis and death rate depend heavily on early detection, tumor type, and available treatments. Brain tumors like glioblastomas, being aggressive, contribute to a relatively high mortality rate compared to other cancers.

The National Cancer Institute presents comprehensive trend graphs on cancer-related mortality, specifically highlighting data on brain tumors(See Figure 2). These visuals provide valuable insights into mortality patterns over time. Cherkasov et al. [12] and Tropsha et al. [13] explored the advancements in QSPR modeling, highlighting the integration of machine learning techniques to enhance the accuracy and generalizability of property predictions across diverse chemical datasets.The foundation of QSPR analysis lies in the principle that molecular structure governs chemical and physical behavior. Various descriptors, including topological indices, electronic properties, and geometric features, serve as numerical representations of molecular characteristics. Advanced statistical techniques and machine learning algorithms are often employed to develop robust QSPR models, enhancing prediction accuracy and reliability[6].

Integrating TIs into QSPR analysis not only enhances the predictive power of the models but also deepens our understanding of the fundamental structure–property correlations. This synergy facilitates the design of new compounds with targeted properties while reducing the need for extensive experimental testing. This article delves

NATIONAL TRENDS IN RATES OF NEW CANCER CASES



AAPC = average annual percent change
*AAPC is significantly different from zero ($p < .05$).

seer.cancer.gov
Source: Annual Report to the Nation

FIGURE 2 Comprehensive trend graphs on cancer-related mortality.

into the methodologies underpinning QSPR analysis, with a focus on the development and application of topological indices, and explores their significant role in advancing research across chemistry, materials science, and related disciplines.

However, selecting the most suitable compound or material often requires evaluating multiple, sometimes conflicting, criteria. Multi-Criteria Decision-Making (MCDM)[32, 33] methods provide a systematic approach to ranking and selecting alternatives based on multiple attributes simultaneously. Several well-known MCDM techniques, such as CRITIC (Criteria Importance Through Intercriteria Correlation), TOPSIS (Technique for Order Preference by Similarity to Ideal Solution), VIKOR (Višekriterijumska Optimizacija i Kompromisno Rešenje), and COPRAS (Complex Proportional Assessment), offer different strategies for analyzing and ranking alternatives based on objective criteria weighting and performance evaluation. Zavadskas et al. [8] analyzed various MCDM techniques, comparing their effectiveness in solving real-world decision-making problems across multiple criteria. Diakoulaki et al. [9] developed the CRITIC method to determine objective criterion weights by evaluating both contrast intensity and intercriteria correlation. Hwang and Yoon [10] introduced the TOPSIS approach, which ranks alternatives by measuring their relative distance to an ideal and an anti-ideal solution. Opricovic and Tzeng [11] proposed the VIKOR method as a tool for multi-criteria optimization, focusing on compromise solutions in complex decision-making scenarios.

- CRITIC determines the importance of criteria by analyzing correlations and contrasts among them, ensuring objective weight assignment.
- TOPSIS ranks alternatives by comparing their distance from an ideal and a negative-ideal solution, favoring options closest to the optimal point.
- VIKOR focuses on compromise solutions by minimizing the maximum regret of alternatives, making it suitable for decision problems involving conflicting objectives.
- COPRAS evaluates alternatives based on proportional dependencies and direct weight assignments, allowing for effective ranking and decision-making.

By integrating QSPR analysis with MCDM techniques[34], researchers can not only predict molecular properties but also make informed decisions about compound selection based on multiple performance criteria. This combination enhances the decision-making process in fields such as drug discovery, materials science, and nanotechnology. This article explores the synergy between QSPR and MCDM methods, highlighting their significance in optimizing molecular design and selecting the best candidates for specific applications. Quantitative Structure–Property Relationship (QSPR) analysis is a widely used computational approach in chemistry and materials science for predicting the physicochemical properties of molecules based on their structural attributes[5]. Todeschini et al. [7] investigated QSPR models to predict molecular properties and biological activities, emphasizing their role in cheminformatics and drug design. Their study focused on the development and validation of molecular descriptors, which serve as numerical representations of chemical structures. They highlighted how QSPR models contribute to various applications, including toxicity prediction, materials science, and environmental chemistry. By establishing mathematical models that correlate molecular descriptors with experimental properties, QSPR enables researchers to design new compounds with desired characteristics without the need for extensive laboratory testing.

2 | MATERIAL AND METHOD

Let G be an undirected graph representing a pharmaceutical compound structure, where the vertex set $V(G)$ corresponds to atoms, and the edge set $E(G)$ represents chemical bonds. An edge joining two vertices u and v is represented as

uv . Let $e = uv \in E(G)$ be an edge of the graph G . In this case, n_u and n_v are defined as the number of vertices that are closer to the respective end vertices u and v of the edge e . Notationally, $n_u = n_u(e) = |N_u(e)|$ and $n_v = n_v(e) = |N_v(e)|$, where $N_u(e) = \{x \in V(G) | d(x, u) < d(x, v)\}$ and $N_v(e) = \{x \in V(G) | d(x, v) < d(x, u)\}$. A vertex $x \in V(G)$ is called an equidistant vertex of $e = uv \in E(G)$ if it is at the same distance from both u and v . It is important to note that the equidistant vertices of e are not included in either n_u or n_v in the graph G . For example, in Figure 3,

- for an edge $e_1 = v_1 v_2$, we have $n_{v_1} = 1$ and $n_{v_2} = 3$ and v_3 is equidistant for e_1 .
- if $e_2 = v_1 v_3$, then $n_{v_1} = 1$ and $n_{v_3} = 2$ and v_2, v_5 are equidistant for e_2 .
- for $e_3 = v_3 v_4$, $n_{v_3} = 1$ and $n_{v_4} = 2$ and v_2, v_5 are equidistant for e_3 .
- if $e_4 = v_2 v_4$, then $n_{v_2} = 3$ and $n_{v_4} = 1$ and v_3 is equidistant for e_4 .
- if $e_5 = v_2 v_5$, then $n_{v_2} = 4$ and $n_{v_5} = 1$.

There are different types of TIs, many of which are motivated by applications in other areas. In this respect, the well-known Wiener index [1] is among the first TIs originating from a chemistry background. Padmakar Khadikar put forward the Padmakar-Ivan (PI) index of graphs in [2]. A vertex version of the PI index is then introduced in Khalifeh et al. [3]. Specifically, we define the vertex PI index of a graph G as follows $PI_v(G) = \sum_{uv \in E(G)} (n_u + n_v)$. The Szeged index [29] is closely related to the Wiener index and defined as $Sz_v(G) = \sum_{uv \in E(G)} n_u n_v$. For more details about these studies, see [4],[5] and [14].

Recently, Gutman and Ashrafi introduced the edge version of the Szeged index [14], which has been computed for various graphs. Notably, edges equidistant from vertices u and v are excluded from consideration. Klavzar [27] provided a comprehensive overview of the cut method, highlighting its applications in chemical graph theory, including the computation of indices such as the Wiener index, Szeged index, hyper-Wiener index, and PI index. John et al. [15] described an orthogonal cut method based on the concept of strongly co-distant edges and utilized it to compute the PI index for certain graphs. In light of these advancements, we undertake the computation of the vertex versions of the KP-polynomial for the proposed carbon structure. This polynomial serve as fundamental tools for establishing various distance-based topological descriptors.

In the literature, a plethora of graph polynomials have been introduced, showcasing their significance in mathematical chemistry. These polynomials, defined in references [20], [21], [22], [23], [24], [25], [26], serve as valuable tools for analyzing graph structures. Graph polynomials are functions of graphs that remain invariant under graph isomorphism, typically characterized by integer coefficients and expressed as polynomials in one or two variables. They act as regular generating functions for coefficient sequences, often enumerating specific subgraphs. One notable polynomial in the realm of distance-based topological indices is the Hosoya polynomial [16],[17],[18] and [19].

A degree-based TIs polynomial dubbed the M-polynomial was recently introduced in 2015 by Klavzar and Deutsch [27]. They demonstrate that its function for degree-based TIs is analogous to the function of the Hosoya polynomial for distance-based TIs.

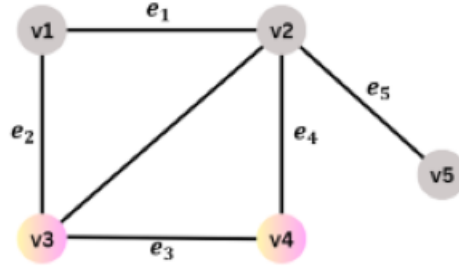


FIGURE 3 The graph G

The M -polynomial of a graph G is defined as

$$M(G, x, y) = \sum_{i \leq j} p_{i,j} x^i y^j, \text{ where } p_{i,j} = |P_{(i,j)}| \text{ and } P_{(i,j)} = \{uv \in E(G) | d_u = i, d_v = j\}.$$

In accordance with polynomials studied previously, Pattabiraman[28] introduce new polynomial called KP -polynomial with respect to distance between vertices in a graph G which is defined as

$$KP_v(G, x, y) = \sum_{i \leq j} \eta_{i,j} x^i y^j, \text{ where } \eta_{i,j} = |E_{(i,j)}| \text{ and } E_{(i,j)} = \{uv \in E(G) | n_u = i, n_v = j\}.$$

The following operations are used to find required results (See Table 1) for this article. $P_x = x \left(\frac{\partial(z(x,y))}{\partial x} \right)$, $P_y = y \left(\frac{\partial(z(x,y))}{\partial y} \right)$, $F_x = \int_0^x \frac{z(t,y)}{t} dt$, $F_y = \int_0^y \frac{z(x,t)}{t} dt$, $\chi_k(z(x,y)) = x^k z(x,y)$ and $J(z(x,y)) = z(x,x)$.

TABLE 1 Formulation of some distance-based TIs using KP-polynomial of a graph G .

TIs	Mathematical expressions	Derivation from $KP_v(G, x, y)$ at $x = 1$ and $y = 1$.
Vertex PI index PI_v	$\sum_{uv \in E(G)} (n_u + n_v)$	$(P_x + P_y)(KP_v(G, x, y))$
Vertex Szeged index Sz_v	$\sum_{uv \in E(G)} n_u n_v$	$(P_x P_y)(KP_v(G, x, y))$
Modified Szeged index Sz^*	$\sum_{uv \in E(G)} \frac{1}{n_u n_v}$	$(F_x F_y)(KP_v(G, x, y))$
Reformulated Szeged index Sz_R	$\sum_{uv \in E(G)} (n_u + n_v) n_u n_v$	$P_x P_y (P_x + P_y)(KP_v(G, x, y))$
F-Szeged index Sz_F	$\sum_{uv \in E(G)} (n_u^2 + n_v^2)$	$(P_x^2 + P_y^2)(KP_v(G, x, y))$
Symmetric deg Szeged index Sz_{SD}	$\sum_{uv \in E(G)} \frac{n_u^2 + n_v^2}{n_u n_v}$	$(P_x F_y + F_x P_y)(KP_v(G, x, y))$
Harmonic Szeged index Sz_H	$\sum_{uv \in E(G)} \frac{2}{n_u + n_v}$	$(2F_x J)(KP_v(G, x, y))$
Inverse sum Szeged index Sz_{ISI}	$\sum_{uv \in E(G)} \frac{n_u n_v}{n_u + n_v}$	$(F_x J P_x P_y)(KP_v(G, x, y))$
Augmented Szeged index Sz_A	$\sum_{uv \in E(G)} \left(\frac{n_u n_v}{n_u + n_v - 2} \right)^3$	$(F_x^3 \psi_{-2} J P_x^3 P_y^3)(KP_v(G, x, y))$
Mostar Szeged index Sz_M	$\sum_{uv \in E(G)} n_u - n_v $	$\left (P_x - P_y) \right (KP_v(G, x, y))$

3 | BRAIN TUMOR TREATMENT DRUGS

This part examines the chemical structures of drugs used to treat brain tumors, emphasizing distance-based Therapeutic Indices (TIs). The discussion includes a Quantitative Structure-Property Relationship (QSPR) analysis of these TIs, demonstrating a significant correlation with the physical properties of the chemical compounds used in brain tumor therapies. Eight brain tumor drugs, depicted in Figure 4, are considered in this analysis:

- **Carmustine**, also known as BCNU (bis-chloroethylnitrosourea), is a chemotherapeutic medication that is commonly used to treat brain malignancies, particularly glioblastoma multiforme, as well as lymphoma and multiple myeloma. It belongs to the nitrosourea class of alkylating drugs, which function by interfering with DNA and RNA synthesis, disrupting cancer cell replication. Carmustine is particularly effective since it can pass the blood-brain barrier and target tumors in the central nervous system. It is frequently delivered intravenously or as an implanted wafer inserted directly into the surgical cavity after tumor removal. Despite its efficiency, carmustine is accompanied with potential side effects such as myelosuppression, pulmonary toxicity, and nausea, which necessitate close monitoring during treatment.
- **Afinitor** (everolimus) is an oral medication used to treat certain types of cancer and tumors, including advanced kidney cancer, breast cancer, and subependymal giant cell astrocytoma (SEGA) associated with tuberous sclerosis complex (TSC). It works by inhibiting the mTOR pathway, which plays a key role in cell growth, proliferation, and survival. By blocking this pathway, Afinitor helps slow tumor growth and spread. Common side effects include fatigue, mouth sores, and gastrointestinal issues, making regular monitoring essential during treatment.
- **Etoposide** is a chemotherapy drug commonly used to treat various cancers, including lung cancer, testicular cancer, and certain types of brain tumors. It works by inhibiting the enzyme topoisomerase II, which is essential for DNA replication and repair, ultimately causing cancer cells to stop dividing and die. Etoposide is available in both oral

and intravenous forms, allowing flexibility in administration. Common side effects include hair loss, low blood cell counts, and nausea, necessitating careful monitoring during treatment.

- **Temozolomide** is an oral chemotherapy drug primarily used to treat brain tumors, including glioblastoma multiforme and anaplastic astrocytoma. It belongs to the alkylating agent class, working by damaging the DNA of cancer cells, which hinders their ability to divide and grow. Its ability to cross the blood-brain barrier makes it particularly effective for central nervous system tumors. Common side effects include nausea, fatigue, and lowered blood cell counts, requiring regular monitoring to manage potential complications during treatment.

- **Procarbazine** is a chemotherapy drug used to treat Hodgkin's lymphoma and certain brain tumors, such as glioblastoma multiforme. It works as an alkylating agent, disrupting DNA synthesis and preventing cancer cells from growing and dividing. Procarbazine is typically administered orally and is often part of combination chemotherapy regimens, such as the PCV protocol (procarbazine, lomustine, and vincristine). Common side effects include nausea, fatigue, and suppressed bone marrow function, making close monitoring necessary during treatment.

- **Tovorafenib** is an investigational oral kinase inhibitor being studied for its potential to treat pediatric brain tumors, particularly low-grade gliomas. It specifically targets aberrant RAF signaling pathways, which are often implicated in tumor growth and progression. By inhibiting these pathways, tovorafenib aims to reduce tumor proliferation and improve patient outcomes. Early clinical trials have shown promise, with manageable side effects such as fatigue, skin rash, and gastrointestinal disturbances, although further studies are needed to establish its safety and efficacy fully.

- **Belzutifan** is an oral medication used to treat certain tumors associated with von Hippel-Lindau (VHL) disease, including renal cell carcinoma, central nervous system hemangioblastomas, and pancreatic neuroendocrine tumors. It works as a hypoxia-inducible factor 2 alpha (HIF-2 α) inhibitor, targeting a pathway involved in tumor growth and survival. By blocking HIF-2 α , belzutifan helps slow tumor progression. Common side effects include fatigue, anemia, and nausea, and patients undergoing treatment require regular monitoring to manage these potential effects.

- **Irinotecan** is a chemotherapy drug used to treat various cancers, including colorectal cancer and certain brain tumors. It functions as a topoisomerase I inhibitor, interfering with the enzyme responsible for unwinding DNA during replication, which leads to DNA damage and cancer cell death. Administered intravenously, irinotecan is often part of combination therapy regimens to enhance its effectiveness. Common side effects include diarrhea, nausea, and reduced blood cell counts, necessitating careful monitoring and supportive care during treatment.

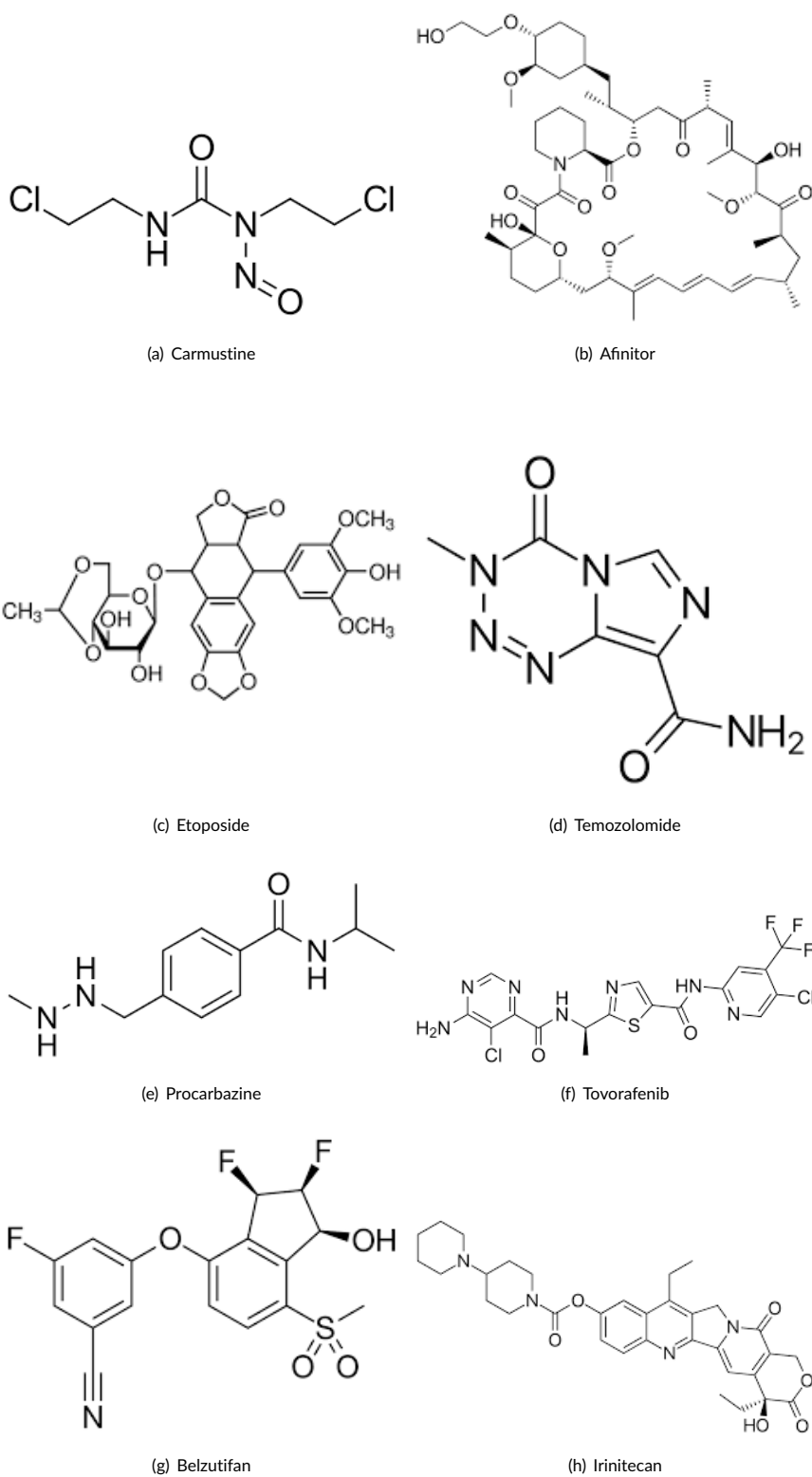


FIGURE 4 Molecular structure of brain tumor treatment drugs.

4 | DETERMINATION OF TIS FOR BRAIN TUMOR TREATMENT DRUGS

This section utilizes distance-based Therapeutic Indices (TIs) to examine the chemical structures of drugs used in the treatment of brain tumor. The analysis includes eight brain tumor drugs, depicted in Figure 4, offering a detailed exploration of their molecular composition.

Theorem 1 Let C be a molecular graph of Carmustine. The KP-polynomial of C is $KP(C, x, y) = 4xy^{11} + 3x^2y^{10} + 2x^3y^9 + x^4y^8 + x^6y^6$.

Proof. From the structure of Carmustine, for $|V(C)| = 12$ and $|E(C)| = 11$, we have the partition of edges into the subsets given as: $|E_{(1,11)}| = 4$, $|E_{(2,10)}| = 3$, $|E_{(3,9)}| = 2$, $|E_{(4,8)}| = 1$ and $|E_{(6,6)}| = 1$. According to the definition of KP-polynomial, we have

$$\begin{aligned} KP(C, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\ &= \sum \eta_{1,11} x y^{11} + \sum \eta_{2,10} x^2 y^{10} + \sum \eta_{3,9} x^3 y^9 + \sum \eta_{4,8} x^4 y^8 + \sum \eta_{6,6} x^6 y^6. \end{aligned}$$

Based on the structure of C , we derive

$$KP(C, x, y) = 4xy^{11} + 3x^2y^{10} + 2x^3y^9 + x^4y^8 + x^6y^6. \quad \square$$

Let $g(x, y) = KP(C, x, y)$. Then by applying Theorem 1, we derive the various distance-based topological descriptors as outlined below.

$$(P_x + P_y)(g(x, y)) = 48xy^{11} + 36x^2y^{10} + 24x^3y^9 + 12x^4y^8 + 12x^6y^6.$$

$$(P_x P_y)(g(x, y)) = 44xy^{11} + 60x^2y^{10} + 54x^3y^9 + 32x^4y^8 + 36x^6y^6.$$

$$F_x F_y(g(x, y)) = \frac{4}{11}xy^{11} + \frac{3}{20}x^2y^{10} + \frac{2}{27}x^3y^9 + \frac{1}{32}x^4y^8 + \frac{1}{36}x^6y^6.$$

$$(P_x + P_y)P_x P_y(g(x, y)) = 528xy^{11} + 720x^2y^{10} + 648x^3y^9 + 384x^4y^8 + 432x^6y^6.$$

$$(P_x^2 + P_y^2)(g(x, y)) = 488xy^{11} + 312x^2y^{10} + 180x^3y^9 + 80x^4y^8 + 72x^6y^6.$$

$$(F_y P_x + F_x P_y)(g(x, y)) = \frac{488}{11}xy^{11} + \frac{78}{5}x^2y^{10} + \frac{20}{3}x^3y^9 + \frac{5}{2}x^4y^8 + \frac{2}{1}x^6y^6.$$

$$F_x J P_y P_x(g(x, y)) = \frac{2}{3}xy^{11} + \frac{1}{2}x^2y^{10} + \frac{1}{3}x^3y^9 + \frac{1}{6}x^4y^8 + \frac{1}{6}x^6y^6.$$

$$F_x J(g(x, y)) = \frac{11}{3}xy^{11} + \frac{5}{1}x^2y^{10} + \frac{9}{2}x^3y^9 + \frac{8}{3}x^4y^8 + \frac{3}{1}x^6y^6.$$

$$F_x^3 Q_{-2} J P_x^3 P_y^3(g(x, y)) = 5.324xy^{11} + 24x^2y^{10} + 39.366x^3y^9 + 32.768x^4y^8 + 46.656x^6y^6.$$

$$|(P_x - P_y)|(g(x, y)) = 160xy^{11} + 72x^2y^{10} + 24x^3y^9 + 4x^4y^8 + 0x^6y^6.$$

Now, by using Table 1, we have,

$$\begin{aligned} PI_v((g(x, y))) &= (P_x + P_y)(g(x, y))|_{(x=1 \text{ and } y=1)} = 132. \\ Sz_v((g(x, y))) &= (P_x P_y)(g(x, y))|_{(x=1 \text{ and } y=1)} = 226. \\ Sz^*((g(x, y))) &= F_x F_y(g(x, y))|_{(x=1 \text{ and } y=1)} = 0.6467. \\ Sz_R((g(x, y))) &= (P_x + P_y)P_x P_y(g(x, y))|_{(x=1 \text{ and } y=1)} = 2712. \\ Sz_F((g(x, y))) &= (P_x^2 + P_y^2)(g(x, y))|_{(x=1 \text{ and } y=1)} = 1132. \\ Sz_{SD}((g(x, y))) &= (F_y P_x + F_x P_y)(g(x, y))|_{(x=1 \text{ and } y=1)} = 72.13. \\ Sz_H((g(x, y))) &= F_x J P_y P_x(g(x, y))|_{(x=1 \text{ and } y=1)} = 1.8333. \\ Sz_{ISI}((g(x, y))) &= F_x J(g(x, y))|_{(x=1 \text{ and } y=1)} = 18.83. \\ Sz_A((g(x, y))) &= F_x^3 Q_{-2} J P_x^3 P_y^3(g(x, y))|_{(x=1 \text{ and } y=1)} = 148.114. \\ Sz_M((g(x, y))) &= |(P_x + P_y)|(g(x, y))|_{(x=1 \text{ and } y=1)} = 260. \end{aligned}$$

Theorem 2 Let A be a molecular graph of Afinitor. The KP-polynomial of A is $KP(A, x, y) = 22xy^{71} + 4x^2y^{70} + 5x^3y^{69} + 3x^4y^{68} + 2x^7y^{65} + 4x^9y^{63} + x^{13}y^{59} + x^{14}y^{58} + x^{16}y^{56} + 3x^{22}y^{47} + 4x^{22}y^{48} + x^{23}y^{31} + 3x^{23}y^{48} + x^{23}y^{45} + x^{23}y^{46} + x^{24}y^{46} + 2x^{24}y^{47} + 6x^{25}y^{45} + x^{26}y^{43} + x^{26}y^{45} + x^{27}y^{41} + 2x^{27}y^{43} + x^{30}y^{40} + x^{30}y^{41} + 3x^{31}y^{40}$.

Proof. From the structure of Afinitor, for $|V(A)| = 72$ and $|E(A)| = 75$, we have the partition of edges into the subsets given as: $|E_{(1,71)}| = 2, |E_{(2,70)}| = 4, |E_{(3,69)}| = 5, |E_{(4,68)}| = 3, |E_{(7,65)}| = 2, |E_{(9,63)}| = 4, |E_{(13,59)}| = 1, |E_{(14,58)}| = 1, |E_{(16,56)}| = 1, |E_{(22,47)}| = 3, |E_{(22,48)}| = 4, |E_{(23,31)}| = 1, |E_{(23,48)}| = 3, |E_{(23,45)}| = 1, |E_{(23,46)}| = 1, |E_{(24,46)}| = 1, |E_{(24,47)}| = 2, |E_{(25,45)}| = 6, |E_{(26,43)}| = 1, |E_{(26,45)}| = 1, |E_{(27,41)}| = 1, |E_{(27,43)}| = 2, |E_{(30,40)}| = 1, |E_{(30,41)}| = 1 and $|E_{(31,40)}| = 3$.$

According to the definition of KP-polynomial, we have

$$\begin{aligned} KP(A, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\ &= \sum \eta_{1,71} x y^{71} + \sum \eta_{2,70} x^2 y^{70} + \sum \eta_{3,69} x^3 y^{69} + \sum \eta_{4,68} x^4 y^{68} \\ &\quad + \sum \eta_{7,65} x^7 y^{65} + \sum \eta_{9,63} x^9 y^{63} + \sum \eta_{13,59} x^{13} y^{59} + \sum \eta_{14,58} x^{14} y^{58} \\ &\quad + \sum \eta_{16,56} x^{16} y^{56} + \sum \eta_{22,47} x^{22} y^{47} + \sum \eta_{22,48} x^{22} y^{48} + \sum \eta_{23,31} x^{23} y^{31} \\ &\quad + \sum \eta_{23,48} x^{23} y^{48} + \sum \eta_{23,45} x^{23} y^{45} + \sum \eta_{23,46} x^{23} y^{46} + \sum \eta_{24,46} x^{24} y^{46} \\ &\quad + \sum \eta_{24,47} x^{24} y^{47} + \sum \eta_{25,45} x^{25} y^{45} + \sum \eta_{26,43} x^{26} y^{43} + \sum \eta_{26,45} x^{26} y^{45} \\ &\quad + \sum \eta_{27,41} x^{27} y^{41} + \sum \eta_{27,43} x^{27} y^{43} + \sum \eta_{30,40} x^{30} y^{40} + \sum \eta_{30,41} x^{30} y^{41} \\ &\quad + \sum \eta_{31,40} x^{31} y^{40}. \end{aligned}$$

Based on the structure of G_2 , we derive

$$\begin{aligned}
 KP(A, x, y) = & 22xy^{71} + 4x^2y^{70} + 5x^3y^{69} + 3x^4y^{68} + 2x^7y^{65} + 4x^9y^{63} \\
 & + x^{13}y^{59} + x^{14}y^{58} + x^{16}y^{56} + 3x^{22}y^{47} + 4x^{22}y^{48} + x^{23}y^{31} \\
 & + 3x^{23}y^{48} + x^{23}y^{45} + x^{23}y^{46} + x^{24}y^{46} + 2x^{24}y^{47} \\
 & + 6x^{25}y^{45} + x^{26}y^{43} + x^{26}y^{45} + x^{27}y^{41} + 2x^{27}y^{43} \\
 & + x^{30}y^{40} + x^{30}y^{41} + 3x^{31}y^{40}.
 \end{aligned}$$

□

Let $f(x, y) = KP(A, x, y)$. Then by applying Theorem 2, we derive the various distance-based topological descriptors as outlined below.

$$\begin{aligned}
 (P_x + P_y)(f(x, y)) = & 1584xy^{71} + 288x^2y^{70} + 360x^3y^{69} + 216x^4y^{68} + 72x^7y^{65} + 144x^9y^{63} \\
 & + 72x^{13}y^{59} + 288x^{14}y^{58} + 72x^{16}y^{56} + 280x^{22}y^{47} + 207x^{22}y^{48} \\
 & + 54x^{23}y^{31} + 213x^{23}y^{48} + 69x^{23}y^{45} + 68x^{23}y^{46} + 70x^{24}y^{46} \\
 & + 142x^{24}y^{47} + 420x^{25}y^{45} + 69x^{26}y^{43} + 71x^{26}y^{45} + 140x^{27}y^{41} \\
 & + 68x^{27}y^{43} + 71x^{30}y^{40} + 70x^{30}y^{41} + 213x^{31}y^{40}.
 \end{aligned}$$

$$\begin{aligned}
 (P_x P_y)(f(x, y)) = & 1562xy^{71} + 560x^2y^{70} + 1035x^3y^{69} + 816x^4y^{68} + 767x^7y^{65} + 910x^9y^{63} \\
 & + 812x^{13}y^{59} + 2268x^{14}y^{58} + 896x^{16}y^{56} + 4224x^{22}y^{47} + 3102x^{22}y^{48} \\
 & + 713x^{23}y^{31} + 3312x^{23}y^{48} + 1058x^{23}y^{45} + 1035x^{23}y^{46} + 1104x^{24}y^{46} \\
 & + 2256x^{24}y^{47} + 6750x^{25}y^{45} + 1118x^{26}y^{43} + 1170x^{26}y^{45} + 2322x^{27}y^{41} \\
 & + 1107x^{27}y^{43} + 1230x^{30}y^{40} + 1200x^{30}y^{41} + 3720x^{31}y^{40}.
 \end{aligned}$$

$$\begin{aligned}
 F_x F_y(f(x, y)) = & \frac{22}{71}xy^{71} + \frac{1}{35}x^2y^{70} + \frac{5}{207}x^3y^{69} + \frac{3}{272}x^4y^{68} + \frac{1}{767}x^7y^{65} + \frac{2}{455}x^9y^{63} \\
 & + \frac{1}{812}x^{13}y^{59} + \frac{4}{567}x^{14}y^{58} + \frac{1}{896}x^{16}y^{56} + \frac{1}{264}x^{22}y^{47} + \frac{3}{1034}x^{22}y^{48} \\
 & + \frac{1}{713}x^{23}y^{31} + \frac{1}{368}x^{23}y^{48} + \frac{1}{1058}x^{23}y^{45} + \frac{1}{1035}x^{23}y^{46} + \frac{1}{1104}x^{24}y^{46} \\
 & + \frac{1}{564}x^{24}y^{47} + \frac{2}{375}x^{25}y^{45} + \frac{1}{1118}x^{26}y^{43} + \frac{1}{1170}x^{26}y^{45} + \frac{2}{1161}x^{27}y^{41} \\
 & + \frac{1}{1107}x^{27}y^{43} + \frac{1}{1230}x^{30}y^{40} + \frac{1}{1200}x^{30}y^{41} + \frac{3}{1240}x^{31}y^{40}.
 \end{aligned}$$

$$\begin{aligned}
(P_x + P_y)P_x P_y(f(x, y)) &= 112464x^7y^{71} + 40320x^2y^{70} + 74520x^3y^{69} + 58752x^4y^{68} \\
&+ 55224x^7y^{65} + 65520x^9y^{63} + 58464x^{13}y^{59} + 163296x^{14}y^{58} \\
&+ 64512x^{16}y^{56} + 295680x^{22}y^{47} + 214038x^{22}y^{48} + 38502x^{23}y^{31} \\
&+ 235152x^{23}y^{48} + 73002x^{23}y^{45} + 70380x^{23}y^{46} + 77280x^{24}y^{46} \\
&+ 160176x^{24}y^{47} + 472500x^{25}y^{45} + 77142x^{26}y^{43} + 83070x^{26}y^{45} \\
&+ 162540x^{27}y^{41} + 75276x^{27}y^{43} + 87330x^{30}y^{40} + 84000x^{30}y^{41} \\
&+ 264120x^{31}y^{40}.
\end{aligned}$$

$$\begin{aligned}
(P_x^2 + P_y^2)(f(x, y)) &= 110924x^7y^{71} + 19616x^2y^{70} + 23850x^3y^{69} + 13920x^4y^{68} \\
&+ 3650x^7y^{65} + 8548x^9y^{63} + 3560x^{13}y^{59} + 16200x^{14}y^{58} \\
&+ 3392x^{16}y^{56} + 11152x^{22}y^{47} + 8079x^{22}y^{48} + 1490x^{23}y^{31} \\
&+ 8499x^{23}y^{48} + 2645x^{23}y^{45} + 2554x^{23}y^{46} + 2692x^{24}y^{46} \\
&+ 5570x^{24}y^{47} + 15900x^{25}y^{45} + 2525x^{26}y^{43} + 2701x^{26}y^{45} \\
&+ 5156x^{27}y^{41} + 2410x^{27}y^{43} + 2581x^{30}y^{40} + 2500x^{30}y^{41} \\
&+ 7683x^{31}y^{40}.
\end{aligned}$$

$$\begin{aligned}
(F_y P_x + F_x P_y)(f(x, y)) &= \frac{110924}{71}xy^{71} + \frac{4904}{35}x^2y^{70} + \frac{2650}{23}x^3y^{69} + \frac{870}{17}x^4y^{68} \\
&+ \frac{3650}{767}x^7y^{65} + \frac{8548}{455}x^9y^{63} + \frac{890}{203}x^{13}y^{59} + \frac{200}{7}x^{14}y^{58} \\
&+ \frac{53}{14}x^{16}y^{56} + \frac{697}{66}x^{22}y^{47} + \frac{8079}{1034}x^{22}y^{48} + \frac{1490}{713}x^{23}y^{31} \\
&+ \frac{2833}{368}x^{23}y^{48} + \frac{5}{2}x^{23}y^{45} + \frac{2554}{1035}x^{23}y^{46} + \frac{673}{276}x^{24}y^{46} \\
&+ \frac{2785}{564}x^{24}y^{47} + \frac{212}{15}x^{25}y^{45} + \frac{2525}{1118}x^{26}y^{43} + \frac{2701}{1170}x^{26}y^{45} \\
&+ \frac{5156}{1161}x^{27}y^{41} + \frac{2410}{1107}x^{27}y^{43} + \frac{2581}{1230}x^{30}y^{40} \\
&+ \frac{25}{12}x^{30}y^{41} + \frac{7683}{1240}x^{31}y^{40}.
\end{aligned}$$

$$\begin{aligned}
F_x J P_y P_x(f(x, y)) &= \frac{11}{18}xy^{71} + \frac{1}{9}x^2y^{70} + \frac{5}{36}x^3y^{69} + \frac{1}{12}x^4y^{68} + \frac{1}{36}x^7y^{65} + \frac{1}{18}x^9y^{63} \\
&+ \frac{1}{36}x^{13}y^{59} + \frac{1}{9}x^{14}y^{58} + \frac{1}{36}x^{16}y^{56} + \frac{4}{35}x^{22}y^{47} + \frac{2}{23}x^{22}y^{48} \\
&+ \frac{1}{27}x^{23}y^{31} + \frac{6}{71}x^{23}y^{48} + \frac{2}{69}x^{23}y^{45} + \frac{1}{34}x^{23}y^{46} + \frac{1}{35}x^{24}y^{46} \\
&+ \frac{4}{71}x^{24}y^{47} + \frac{6}{35}x^{25}y^{45} + \frac{2}{69}x^{26}y^{43} + \frac{2}{71}x^{26}y^{45} + \frac{2}{35}x^{27}y^{41} \\
&+ \frac{1}{34}x^{27}y^{43} + \frac{2}{71}x^{30}y^{40} + \frac{1}{35}x^{30}y^{41} + \frac{6}{71}x^{31}y^{40}.
\end{aligned}$$

$$\begin{aligned}
F_x J(f(x, y)) = & \frac{781}{36} x y^{71} + \frac{70}{9} x^2 y^{70} + \frac{115}{8} x^3 y^{69} + \frac{34}{3} x^4 y^{68} + \frac{767}{72} x^7 y^{65} + \frac{455}{36} x^9 y^{63} \\
& + \frac{203}{18} x^{13} y^{59} + \frac{63}{2} x^{14} y^{58} + \frac{112}{9} x^{16} y^{56} + \frac{2112}{35} x^{22} y^{47} + \frac{1034}{23} x^{22} y^{48} \\
& + \frac{713}{54} x^{23} y^{31} + \frac{3312}{71} x^{23} y^{48} + \frac{46}{3} x^{23} y^{45} + \frac{1035}{68} x^{23} y^{46} + \frac{552}{35} x^{24} y^{46} \\
& + \frac{2256}{71} x^{24} y^{47} + \frac{675}{7} x^{25} y^{45} + \frac{118}{69} x^{26} y^{43} + \frac{1170}{71} x^{26} y^{45} + \frac{1161}{35} x^{27} y^{41} \\
& + \frac{1107}{68} x^{27} y^{43} + \frac{1230}{71} x^{30} y^{40} + \frac{120}{7} x^{30} y^{41} + \frac{3720}{71} x^{31} y^{40}.
\end{aligned}$$

$$\begin{aligned}
F_x^3 Q_{-2} J P_x^3 P_y^3(f(x, y)) = & 22.95 x y^{71} + 32 x^2 y^{70} + 129.29 x^3 y^{69} + 176.00 x^4 y^{68} \\
& + 1315.50 x^7 y^{65} + 549.25 x^9 y^{63} + 1560.89 x^{13} y^{59} \\
& + 2125.76 x^{14} y^{58} + 2097.15 x^{16} y^{56} + 14980.45 x^{22} y^{47} \\
& + 11027.02 x^{22} y^{48} + 2577.85 x^{23} y^{31} + 12288 x^{23} y^{48} \\
& + 3937.60 x^{23} y^{45} + 3856.46 x^{23} y^{46} + 4279.37 x^{24} y^{46} \\
& + 8737.95 x^{24} y^{47} + 27169.52 x^{25} y^{45} + 4646.23 x^{26} y^{43} \\
& + 4875.40 x^{26} y^{45} + 9954.05 x^{27} y^{41} + 4718.57 x^{27} y^{43} \\
& + 5664.58 x^{30} y^{40} + 5495.62 x^{30} y^{41} + 17411.61 x^{31} y^{40}.
\end{aligned}$$

$$\begin{aligned}
|(P_x - P_y)|(f(x, y)) = & 33880 x y^{71} + 1088 x^2 y^{70} + 1650 x^3 y^{69} + 576 x^4 y^{68} + 46 x^7 y^{65} \\
& + 232 x^9 y^{63} + 44 x^{13} y^{59} + 864 x^{14} y^{58} + 40 x^{16} y^{56} + 416 x^{22} y^{47} \\
& + 225 x^{22} y^{48} + 8 x^{23} y^{31} + 225 x^{23} y^{48} + 23 x^{23} y^{45} + 22 x^{23} y^{46} \\
& + 22 x^{24} y^{46} + 92 x^{24} y^{47} + 720 x^{25} y^{45} + 17 x^{26} y^{43} + 19 x^{26} y^{45} \\
& + 64 x^{27} y^{41} + 14 x^{27} y^{43} + 11 x^{30} y^{40} + 10 x^{30} y^{41} + 81 x^{31} y^{40}.
\end{aligned}$$

Now, by using Table 1, we have,

$$\begin{aligned}
PI_V(f(x, y)) &= (P_x + P_y)(f(x, y))|_{(x=1 \text{ and } y=1)} = 5321. \\
SZ_V(f(x, y)) &= (P_x P_y)(f(x, y))|_{(x=1 \text{ and } y=1)} = 45047. \\
SZ^*(f(x, y)) &= F_x F_y(f(x, y))|_{(x=1 \text{ and } y=1)} = 0.4179. \\
SZ_R(f(x, y)) &= (P_x + P_y) P_x P_y(f(x, y))|_{(x=1 \text{ and } y=1)} = 3163260. \\
SZ_F(f(x, y)) &= (P_x^2 + P_y^2)(f(x, y))|_{(x=1 \text{ and } y=1)} = 287797. \\
SZ_{SD}(f(x, y)) &= (F_y P_x + F_x P_y)(f(x, y))|_{(x=1 \text{ and } y=1)} = 2005.31. \\
SZ_H(f(x, y)) &= F_x J P_y P_x(f(x, y))|_{(x=1 \text{ and } y=1)} = 2.1169. \\
SZ_{ISI}(f(x, y)) &= F_x J(f(x, y))|_{(x=1 \text{ and } y=1)} = 642.3678. \\
SZ_A(f(x, y)) &= F_x^3 Q_{-2} J P_x^3 P_y^3(KP(A, x, y))|_{(x=1 \text{ and } y=1)} = 149629.2. \\
SZ_M(f(x, y)) &= |(P_x + P_y)|(f(x, y))|_{(x=1 \text{ and } y=1)} = 40389.
\end{aligned}$$

Theorem 3 Let E_t be a molecular graph of Etoposide. The KP-polynomial of E_t is $KP(E_t, x, y) = 9xy^{43} + 2x^2y^{21} + x^2y^{23} + x^2y^{40} + 2x^2y^{41} + 2x^2y^{42} + x^3y^{23} + x^3y^{40} + 4x^4y^{40} + 8x^6y^{38} + 7x^8y^{36} + 2x^9y^{35} + x^{11}y^{33} + x^{13}y^{31} + x^{14}y^{30} + 3x^{20}y^{22} + 2x^{20}y^{24} + 2x^{21}y^{23}$.

Proof. From the structure of Etoposide, for $|V(E_t)| = 44$ and $|E(E_t)| = 50$, we have the partition of edges into the subsets given as: $|E_{(1,43)}| = 9$, $|E_{(2,21)}| = 2$, $|E_{(2,23)}| = 1$, $|E_{(2,40)}| = 1$, $|E_{(2,41)}| = 2$, $|E_{(2,42)}| = 2$, $|E_{(3,23)}| = 1$, $|E_{(3,40)}| = 1$, $|E_{(4,40)}| = 4$, $|E_{(6,38)}| = 8$, $|E_{(8,36)}| = 7$, $|E_{(9,35)}| = 2$, $|E_{(11,33)}| = 1$, $|E_{(13,31)}| = 1$, $|E_{(14,30)}| = 1$, $|E_{(20,22)}| = 3$, $|E_{(20,24)}| = 2$ and $|E_{(21,23)}| = 2$. According to the definition of KP-polynomial, we have

$$\begin{aligned} KP(E_t, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\ &= \sum \eta_{1,43} x y^{43} + \sum \eta_{2,21} x^2 y^{21} + \sum \eta_{2,23} x^2 y^{23} + \sum \eta_{2,40} x^2 y^{40} \\ &\quad + \sum \eta_{2,41} x^2 y^{41} + \sum \eta_{2,42} x^2 y^{42} + \sum \eta_{3,23} x^3 y^{23} + \sum \eta_{3,40} x^3 y^{40} \\ &\quad + \sum \eta_{4,40} x^4 y^{40} + \sum \eta_{6,38} x^6 y^{38} + \sum \eta_{8,36} x^8 y^{36} + \sum \eta_{9,35} x^9 y^{35} \\ &\quad + \sum \eta_{11,33} x^{11} y^{33} + \sum \eta_{13,31} x^{13} y^{31} + \sum \eta_{14,30} x^{14} y^{30} \\ &\quad + \sum \eta_{20,22} x^{20} y^{22} + \sum \eta_{20,24} x^{20} y^{24} + \sum \eta_{21,23} x^{21} y^{23}. \end{aligned}$$

Based on the structure of E_t , we derive

$$\begin{aligned} KP(E_t, x, y) &= 9xy^{43} + 2x^2y^{21} + x^2y^{23} + x^2y^{40} + 2x^2y^{41} + 2x^2y^{42} + x^3y^{23} \\ &\quad + x^3y^{40} + 4x^4y^{40} + 8x^6y^{38} + 7x^8y^{36} + 2x^9y^{35} + x^{11}y^{33} \\ &\quad + x^{13}y^{31} + x^{14}y^{30} + 3x^{20}y^{22} + 2x^{20}y^{24} + 2x^{21}y^{23}. \end{aligned} \quad \square$$

Theorem 4 Let T_e be a molecular graph of Temozolomide. The KP-polynomial of T_e is $KP(T_e, x, y) = 4xy^{13} + x^2y^7 + x^2y^8 + x^3y^{11} + x^4y^9 + 2x^4y^{10} + x^5y^6 + x^5y^8 + 2x^5y^9 + x^6y^7$.

Proof. From the structure of Temozolomide, for $|V(T_e)| = 14$ and $|E(T_e)| = 15$, we have the partition of edges into the subsets given as: $|E_{(1,13)}| = 4$, $|E_{(2,7)}| = 1$, $|E_{(2,8)}| = 1$, $|E_{(3,11)}| = 1$, $|E_{(4,9)}| = 1$, $|E_{(4,10)}| = 2$, $|E_{(5,6)}| = 1$, $|E_{(5,8)}| = 1$, $|E_{(5,9)}| = 2$ and $|E_{(6,7)}| = 1$. According to the definition of KP-polynomial, we have

$$\begin{aligned} KP(T_e, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\ &= \sum \eta_{1,13} x y^{13} + \sum \eta_{2,7} x^2 y^7 + \sum \eta_{2,8} x^2 y^8 + \sum \eta_{3,11} x^3 y^{11} + \sum \eta_{4,9} x^4 y^9 \\ &\quad + \sum \eta_{4,10} x^4 y^{10} + \sum \eta_{5,6} x^5 y^6 + \sum \eta_{5,8} x^5 y^8 + \sum \eta_{5,9} x^5 y^9 + \sum \eta_{6,7} x^6 y^7. \end{aligned}$$

Based on the structure of T_e , we derive

$$KP(T_e, x, y) = 4xy^{13} + x^2y^7 + x^2y^8 + x^3y^{11} + x^4y^9 + 2x^4y^{10} + x^5y^6 + x^5y^8 + 2x^5y^9 + x^6y^7. \quad \square$$

Theorem 5 Let P be a molecular graph of Procarbazine. The KP-polynomial of P is $KP(P, x, y) = 4xy^{15} + x^2y^{14} + 2x^3y^{13} + 2x^4y^{12} + x^6y^{10} + 6x^7y^9$.

Proof. From the structure of Procarbazine, for $|V(P)| = 16$ and $|E(P)| = 16$, we have the partition of edges into the subsets given as: $|E_{(1,15)}| = 4$, $|E_{(2,14)}| = 1$, $|E_{(3,13)}| = 2$, $|E_{(4,12)}| = 2$, $|E_{(6,10)}| = 1$ and $|E_{(7,9)}| = 6$. According to the

definition of KP-polynomial, we have

$$\begin{aligned} KP(P, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\ &= \sum \eta_{1,15} x y^{15} + \sum \eta_{2,14} x^2 y^{14} + \sum \eta_{3,13} x^3 y^{13} + \sum \eta_{4,12} x^4 y^{12} + \sum \eta_{6,10} x^6 y^{10} \\ &\quad + \sum \eta_{7,9} x^7 y^9. \end{aligned}$$

Based on the structure of P , we derive

$$KP(P, x, y) = 4xy^{15} + x^2y^{14} + 2x^3y^{13} + 2x^4y^{12} + x^6y^{10} + 6x^7y^9. \quad \square$$

Theorem 6 Let T_o be a molecular graph of Tovorafenib. The KP-polynomial of T_o is $KP(T_o, x, y) = 9xy^{31} + x^2y^{15} + x^2y^{16} + 3x^3y^{29} + 4x^4y^{28} + 2x^5y^{27} + 5x^8y^{24} + x^{10}y^{22} + 2x^{11}y^{21} + x^{12}y^{20} + x^{13}y^{19} + x^{14}y^{18} + 3x^{15}y^{16}$.

Proof. From the structure of Tovorafenib, for $|V(T_o)| = 32$ and $|E(T_o)| = 34$, we have the partition of edges into the subsets given as: $|E_{(1,31)}| = 9, |E_{(2,15)}| = 1, |E_{(2,16)}| = 1, |E_{(3,29)}| = 3, |E_{(4,28)}| = 4, |E_{(5,27)}| = 2, |E_{(8,24)}| = 5, |E_{(10,22)}| = 1, |E_{(11,21)}| = 2, |E_{(12,20)}| = 1, |E_{(13,19)}| = 1, |E_{(15,16)}| = 3$ and $|E_{(14,18)}| = 1$. According to the definition of KP-polynomial, we have

$$\begin{aligned} KP(T_o, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\ &= \sum \eta_{1,31} x y^{31} + \sum \eta_{2,15} x^2 y^{15} + \sum \eta_{2,16} x^2 y^{16} + \sum \eta_{3,29} x^3 y^{29} \\ &\quad + \sum \eta_{4,28} x^4 y^{28} + \sum \eta_{5,27} x^5 y^{27} + \sum \eta_{8,24} x^8 y^{24} + \sum \eta_{10,22} x^{10} y^{22} \\ &\quad + \sum \eta_{11,21} x^{11} y^{21} + \sum \eta_{12,20} x^{12} y^{20} + \sum \eta_{13,19} x^{13} y^{19} \\ &\quad + \sum \eta_{14,18} x^{14} y^{18} + \sum \eta_{15,16} x^{15} y^{16}. \end{aligned}$$

Based on the structure of T_o , we derive

$$\begin{aligned} KP(T_o, x, y) &= 9xy^{31} + x^2y^{15} + x^2y^{16} + 3x^3y^{29} + 4x^4y^{28} + 2x^5y^{27} + 5x^8y^{24} \\ &\quad + x^{10}y^{22} + 2x^{11}y^{21} + x^{12}y^{20} + x^{13}y^{19} + x^{14}y^{18} + 3x^{15}y^{16}. \quad \square \end{aligned}$$

Theorem 7 Let B be a molecular graph of Belzutifan. The KP-polynomial of B is $KP(B, x, y) = 8xy^{24} + x^2y^{13} + x^4y^9 + x^4y^{14} + 3x^4y^{19} + 5x^4y^{21} + 2x^5y^{20} + 2x^7y^{18} + x^8y^{17} + x^9y^{14} + x^9y^{16} + x^{12}y^{13}$.

Proof. From the structure of Belzutifan, for $|V(B)| = 25$ and $|E(B)| = 27$, we have the partition of edges into the subsets given as: $|E_{(1,24)}| = 8, |E_{(2,13)}| = 1, |E_{(4,21)}| = 5, |E_{(5,20)}| = 2, |E_{(4,9)}| = 1, |E_{(4,14)}| = 1, |E_{(4,19)}| = 3, |E_{(7,18)}| = 2, |E_{(8,17)}| = 1, |E_{(9,14)}| = 1, |E_{(9,16)}| = 1$ and $|E_{(12,13)}| = 1$. According to the definition of KP-polynomial, we

have

$$\begin{aligned}
 KP(B, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\
 &= \sum \eta_{1,24} x y^{24} + \sum \eta_{2,13} x^2 y^{13} + \sum \eta_{4,9} x^4 y^9 + \sum \eta_{4,14} x^4 y^{14} + \sum \eta_{4,19} x^4 y^{19} \\
 &\quad + \sum \eta_{4,21} x^4 y^{21} + \sum \eta_{5,20} x^5 y^{20} + \sum \eta_{7,18} x^7 y^{18} + \sum \eta_{8,17} x^8 y^{17} \\
 &\quad + \sum \eta_{9,14} x^9 y^{14} + \sum \eta_{9,16} x^9 y^{16} + \sum \eta_{12,13} x^{12} y^{13}.
 \end{aligned}$$

Based on the structure of B , we derive

$$\begin{aligned}
 KP(B, x, y) &= 8xy^{24} + x^2y^{13} + x^4y^9 + x^4y^{14} + 3x^4y^{19} + 5x^4y^{21} + 2x^5y^{20} + 2x^7y^{18} + x^8y^{17} \\
 &\quad + x^9y^{14} + x^9y^{16} + x^{12}y^{13}. \quad \square
 \end{aligned}$$

Theorem 8 Let I be a molecular graph of Irinotecan. The KP-polynomial of I is $KP(I, x, y) = 6xy^{42} + 2x^2y^{41} + 6x^3y^{40} + 2x^4y^{39} + x^6y^{37} + 3x^7y^{14} + x^7y^{28} + 2x^7y^{36} + 6x^9y^{34} + 2x^{11}y^{32} + 3x^{12}y^{31} + 3x^{14}y^{23} + x^{14}y^{29} + x^{15}y^{23} + x^{15}y^{27} + x^{15}y^{28} + 4x^{18}y^{25} + 4x^{21}y^{22}$.

Proof. From the structure of Irinotecan, for $|V(I)| = 43$ and $|E(I)| = 49$, we have the partition of edges into the subsets given as: $|E_{(1,42)}| = 6$, $|E_{(2,41)}| = 2$, $|E_{(3,40)}| = 6$, $|E_{(4,39)}| = 2$, $|E_{(6,37)}| = 1$, $|E_{(7,14)}| = 3$, $|E_{(7,28)}| = 1$, $|E_{(7,36)}| = 2$, $|E_{(9,34)}| = 6$, $|E_{(11,32)}| = 2$, $|E_{(12,31)}| = 3$, $|E_{(15,23)}| = 1$, $|E_{(15,27)}| = 1$, $|E_{(15,28)}| = 1$, $|E_{(14,29)}| = 1$, $|E_{(14,23)}| = 3$, $|E_{(18,25)}| = 4$ and $|E_{(21,22)}| = 4$. According to the definition of KP-polynomial, we have

$$\begin{aligned}
 KP(I, x, y) &= \sum_{i \leq j} \eta_{ij} x^i y^j \\
 &= \sum \eta_{1,42} x y^{42} + \sum \eta_{2,41} x^2 y^{41} + \sum \eta_{3,40} x^3 y^{40} + \sum \eta_{4,39} x^4 y^{39} \\
 &\quad + \sum \eta_{6,37} x^6 y^{37} + \sum \eta_{7,14} x^7 y^{14} + \sum \eta_{7,28} x^7 y^{28} + \sum \eta_{7,36} x^7 y^{36} \\
 &\quad + \sum \eta_{9,34} x^9 y^{34} + \sum \eta_{11,32} x^{11} y^{32} + \sum \eta_{12,31} x^{12} y^{31} + \sum \eta_{14,23} x^{14} y^{23} \\
 &\quad + \sum \eta_{14,29} x^{14} y^{29} + \sum \eta_{15,23} x^{15} y^{23} + \sum \eta_{15,27} x^{15} y^{27} \\
 &\quad + \sum \eta_{15,28} x^{15} y^{28} + \sum \eta_{18,25} x^{18} y^{25} + \sum \eta_{21,22} x^{21} y^{22}.
 \end{aligned}$$

Based on the structure of I , we derive

$$\begin{aligned}
 KP(I, x, y) &= 6xy^{42} + 2x^2y^{41} + 6x^3y^{40} + 2x^4y^{39} + x^6y^{37} + 3x^7y^{14} + x^7y^{28} + 2x^7y^{36} \\
 &\quad + 6x^9y^{34} + 2x^{11}y^{32} + 3x^{12}y^{31} + 3x^{14}y^{23} + x^{14}y^{29} + x^{15}y^{23} + x^{15}y^{27} \\
 &\quad + x^{15}y^{28} + 4x^{18}y^{25} + 4x^{21}y^{22}. \quad \square
 \end{aligned}$$

The surface depiction of the KP-polynomials given in the Theorems 1 to Theorem 8 for the molecular graphs of proposed drugs are illustrated in Figure 5. From these, it is evident that the KP-polynomial exhibits varying behavior as the structure changes. As presented in Table 2, various distance-based TIs are derived from Theorems 1-8 and Table 1.

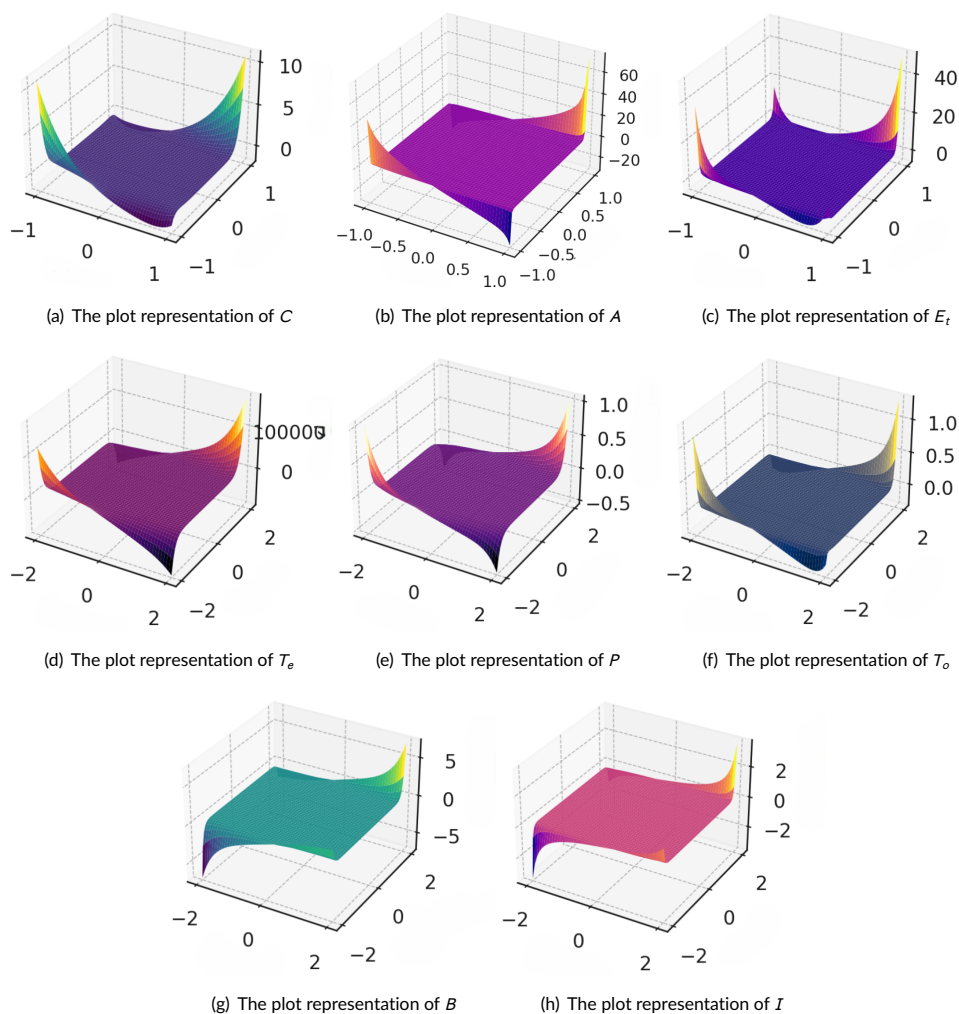


FIGURE 5 The plot representation of the eight molecular graphs.

5 | QUALITY TESTING ANALYSIS

This section examined the application of TIs in the treatment of brain tumors, highlighting the strong correlation between Quantitative Structure-Property Relationships (QSPR) and TIs based on their computational results. The medications listed in Table 2 play a significant role in the mathematical analysis related to brain tumor treatment.

The primary focus of this study is on eight TIs utilized in QSPR modeling. These eight drugs, used for treating brain tumors, exhibit varying properties such as boiling point (BP), flash point (FP), enthalpy of vaporization (EOV), molar mass (MM), molar refractivity (MR), polarizability (PZ), and molar volume (MV). The values of these properties are given in the Table 3. Equation (1) was employed to evaluate the linear regression models applied to these medications. By

TABLE 2 The values of distance based TIs for the proposed drugs.

Drug Name	PI_v	Sz_v	Sz_F	Sz_R	Sz^*	Sz_{SD}	Sz_{ISI}	Sz_H	Sz_A	Sz_M
Carmustine	132	226	0.6467	2712	1132	72.13	1.8333	18.83333	148.114	260
Afinitor	5321	45047	0.4179	3163260	287797	2005.31	2.1169	642.3678	149629.2	40389
Etoposide	2110	10660	0.4757	462076	69146	700.97	2.4336	247.5613	17273.4	8108
Temozolomide	195	433	0.6763	5720	1707	83.77	2.3513	33.174	397.26	261
Procarbazine	256	700	0.5072	11200	2696	97.85	2	43.75	762.23	384
Tovorafenib	1056	4421	0.5038	139854	24350	400.78	2.2348	140.487	6470.90	3642
Belzutifan	638	1972	0.5809	47508	11390	271.11	2.3461	83.00497	1954.18	2184
Irinotecan	2009	12510	0.3487	521968	58803	494.52	2.4657	304.0578	24840.12	4747

TABLE 3 Chemical properties of the proposed brain tumor drugs.

Drug Name	BP	FP	EOV	MOV	MM	MR	PZ	MV
Carmustine	309.6	141	63.8	214.05	213.007187	46.6	18.5	146.4
Afinitor	998.7	557.8	165.1	958.224	957.58136	257.7	102.2	811.2
Etoposide	798.1	263.6	121.7	588.557	588.184265	140.1	55.5	378.5
Temozolomide	526.6	272.3	80.1	194.151	194.055222	45.6	18.1	98.4
Procarbazine	384.6	148.9	63.3	221.299	221.152817	65.8	26.1	213.6
Tovorafenib	585.3	307.8	87.4	506.289	505.010223	113.3	44.9	308.7
Belzutifan	505.8	259.7	81.7	383.342	383.043915	84.9	33.6	244.7
Irinotecan	873.4	482	133	586.678	586.279114	159.1	63.1	416.8

TABLE 4 The value of statistical parameters of PI_v index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
EOV	0.0198	70.5303	0.9445	7.0395	0.0004
MOV	0.1443	245.2067	0.9641	8.8958	0.0001
MM	0.1443	244.7943	0.9641	8.9072	0.0001
MR	0.0402	55.3131	0.9799	12.0262	0
PZ	0.0159	21.9223	0.98	12.0523	0
MV	0.1265	141.9776	0.9866	14.8284	0

TABLE 5 The value of statistical parameters of S_z index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
<i>MOV</i>	0.0158	306.1726	0.918	5.6708	0.0013
<i>MM</i>	0.0158	305.7546	0.9181	5.6729	0.0013
<i>MR</i>	0.0045	71.662	0.9468	7.2069	0.0004
<i>PZ</i>	0.0018	28.4032	0.947	7.2232	0.0004
<i>MV</i>	0.0144	190.9617	0.9713	9.9973	0.0001

TABLE 6 The value of statistical parameters of S_{z_R} index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
<i>MR</i>	0.0001	81.5927	0.9024	5.1298	0.0022
<i>PZ</i>	0	32.3411	0.9027	5.1382	0.0021
<i>MV</i>	0.0002	220.923	0.9427	6.9181	0.0005

TABLE 7 The value of statistical parameters of S_{z_F} index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
<i>MOV</i>	0.0024	317.2849	0.9049	5.2093	0.002
<i>MM</i>	0.0024	316.8716	0.905	5.2097	0.002
<i>MR</i>	0.0007	74.9355	0.9301	6.2036	0.0008
<i>PZ</i>	0.0003	29.7019	0.9303	6.2135	0.0008
<i>MV</i>	0.0022	200.5141	0.9614	8.5571	0.0001

TABLE 8 The value of statistical parameters of $S_{z_{SD}}$ index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
<i>MOV</i>	0.3815	259.8128	0.9382	6.6406	0.0006
<i>MM</i>	0.3814	259.4216	0.9382	6.639	0.0006
<i>MR</i>	0.1056	59.6513	0.9488	7.3567	0.0003
<i>PZ</i>	0.0419	23.643	0.9489	7.3622	0.0003
<i>MV</i>	0.3388	152.5236	0.9727	10.2651	0

TABLE 9 The value of statistical parameters of $ISIZ$ index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
<i>BP</i>	1.0532	423.5494	0.9115	5.4293	0.0016
<i>EOV</i>	0.1668	67.9647	0.9609	8.5018	0.0001
<i>MOV</i>	1.1989	229.7986	0.9668	9.2728	0.0001
<i>MM</i>	1.1988	229.3738	0.967	9.2951	0.0001
<i>MR</i>	0.3357	50.6375	0.9887	16.138	0
<i>PZ</i>	0.1331	20.0672	0.9888	16.228	0
<i>MV</i>	1.0489	128.8803	0.9874	15.2616	0

TABLE 10 The value of statistical parameters of Sz_A index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
<i>MV</i>	0.0041	223.9005	0.9381	6.6337	0.0006

TABLE 11 The value of statistical parameters of Sz_M index against the properties of the drugs.

Property	α	β	r	t -statistic	p -value
<i>MV</i>	0.0155	210.977	0.9412	6.8261	0.0005

leveraging this equation, a diverse range of linear models can be developed to identify additional TIs, as demonstrated below.

$$Z = \alpha + \beta_1 X_1 \quad (1)$$

In this analysis, Z represents the response or dependent variable, while the regression model constant and the coefficients for individual descriptors are used alongside X , the independent variable. Parameters such as the correlation coefficient (r), t -statistic [35] and significance value (p) were utilized to evaluate the reliability of the results. Regression equations were organized into tables to assess how well the models fit the data.

During the calculation of reducible indices, a strong connection was observed, with correlations ranging from 0.7 to 0.9. The square of the correlation coefficient (r^2) indicates the strength of the relationship between two variables, whether positive or negative. The highest adjusted r^2 value signifies the best model fit, providing the most accurate equation, demonstrating a strong positive correlation between these variables.

This analysis enhances our understanding of the relationship between the properties of brain tumor drugs and their effectiveness. It emphasizes the interconnectedness of various factors in the context of brain tumor treatments. Finally, Figure 6 provides a graphical comparison of correlation coefficients between drug properties and reducible indices.

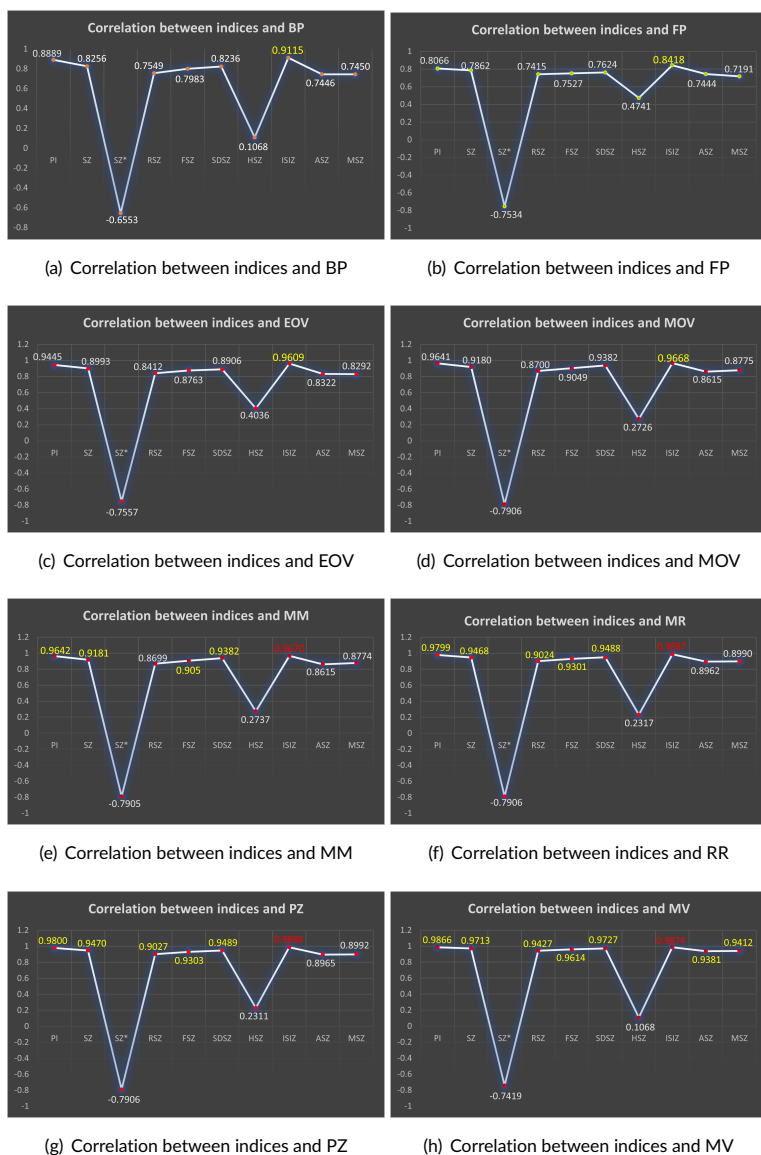


FIGURE 6 Visual representation of the correlation (R values) between properties and topological indices.

Based on our analysis of the proposed topological indices (TIs) presented in Tables 4 to 11 and Figure 6, we have observed the following:

- Boiling Point (BP):** The property BP can be effectively predicted using the PI_v index, as its corresponding correlation coefficient (r) is 0.9445. This indicates that PI_v is the most suitable index for BP prediction.

- Enthalpy of Vaporization (EOV):** Both PI_v and $ISIZ$ indices can be used to predict EOV, with correlation coefficients of 0.9445 and 0.9609, respectively. Among these, $ISIZ$ is the most accurate predictor for EOV.

•**Molecular Volume (MOV):** The indices PI_V , Sz , Sz_F , Sz_{SD} , and $ISIZ$ show strong correlation with MOV, with respective r values of 0.9641, 0.918, 0.9049, 0.9382, and 0.9668. Among these, $ISIZ$ is the most reliable for predicting MOV.

•**Molar Mass (MM):** The indices PI_V , Sz , Sz_F , Sz_{SD} , and $ISIZ$ also serve as good predictors for MM, with correlation values of 0.9642, 0.9181, 0.905, 0.9382, and 0.967, respectively. $ISIZ$ emerges as the most effective index for predicting MM.

•**Molar Refractivity (MR):** The indices PI_V , Sz , Sz_R , Sz_F , Sz_{SD} , and $ISIZ$ correlate with MR, yielding r values of 0.9799, 0.9468, 0.9024, 0.9301, 0.9488, and 0.9887, respectively. Among these, $ISIZ$ is the most suitable for MR prediction.

•**Polarizability (PZ):** The indices PI_V , Sz , Sz_R , Sz_F , Sz_{SD} , and $ISIZ$ also demonstrate strong correlation with PZ, with respective r values of 0.98, 0.947, 0.9027, 0.9303, 0.9489, and 0.9888. The most appropriate predictor for PZ is $ISIZ$.

•**Molar Volume (MV):** The indices PI_V , Sz , Sz_R , Sz_F , Sz_{SD} , $ISIZ$, Sz_A , and Sz_M show significant correlation with MV, with r values of 0.9866, 0.9713, 0.9427, 0.9614, 0.9727, 0.9874, 0.9381, and 0.9412, respectively. Among these, $ISIZ$ is identified as the most reliable predictor for MV.

Overall, our findings suggest that the $ISIZ$ index consistently provides the highest predictive accuracy for multiple properties, making it the most effective topological index among those analyzed.

6 | MULTICRITERIA DECISION MAKING

In the next subsections, we will look at how to apply the MCDM technique. The presentation will cover the processes needed to do mathematical calculations for various strategies.

6.1 | CRITIC method

Critical Requirements: One method of Multi-Criteria Decision Making (MCDM) involves using intercriteria correlation, which analyzes the relationships between different criteria to assign their respective weights[38]. This approach aims to minimize reliance on subjective judgment by determining the weight of each criterion based on its correlation with others. The following outlines the steps involved in the CRITIC method.

Step 1: The first step involves constructing a decision matrix X , where the criteria are arranged in columns, and the alternatives are listed in rows, as shown in Table 2.

$$X = [x_{iq}]_{m \times k} = \begin{bmatrix} x_{11} & x_{12} & \dots & x_{1k} \\ x_{21} & x_{22} & \dots & x_{2k} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \dots & x_{mk} \end{bmatrix}$$

where $i = 1, \dots, k$ and $q = 1, \dots, m$.

The effectiveness value of the i th alternative for the q th criterion is denoted by X_{iq} .

Step 2: The following equation is used to normalize the decision matrix. The resulting normalized matrix is

presented in Table 12.

$$x'_{iq} = \frac{x_{iq} - \min(x_q)}{\max(x_q) - \min(x_q)} \quad (2)$$

Where: - x_{iq} is the value of alternative i for criterion q , - $\min(x_q)$ and $\max(x_q)$ are the minimum and maximum values of criterion q , respectively.

TABLE 12 Normalised decision matrix for CRITIC method.

Drug Name	PI_V	Sz_V	Sz_F	Sz_R	Sz^*	Sz_{SD}	Sz_{ISI}	Sz_H	Sz_A	Sz_M
Carmustine	0	0	0.9096	0	0	0	0	0	0	0
Afinitor	1	1	0.2112	1	1	1	0.448	1	1	1
Etoposide	0.381	0.233	0.3877	0.145	0.2373	0.3253	0.949	0.367	0.1146	0.1956
Temozolomide	0.012	0.005	1	1E-03	0.002	0.006	0.819	0.023	0.0017	2E-05
Procarbazine	0.024	0.011	0.4838	0.003	0.0055	0.0133	0.264	0.04	0.0041	0.0031
Tovorafenib	0.178	0.094	0.4734	0.043	0.081	0.17	0.635	0.195	0.0423	0.0843
Belzutifan	0.098	0.039	0.7088	0.014	0.0358	0.1029	0.811	0.103	0.0121	0.0479
Irinotecan	0.362	0.274	0	0.164	0.2012	0.2185	1	0.457	0.1652	0.1118

TABLE 13 Weight calculation for each criterion by using CRITIC method.

Topological indices	$\sum(1 - \sigma_{qj})$	σ_q	C_q	W_q
PI_V	2.7032	0.3360	0.9084	0.0596
Sz_V	2.6519	0.3375	0.8951	0.0587
Sz_F	13.9731	0.3390	4.7372	0.3107
Sz_R	2.7033	0.3413	0.9226	0.0605
Sz^*	2.6369	0.3379	0.8911	0.0584
Sz_{SD}	2.6380	0.3321	0.8760	0.0575
Sz_{ISI}	9.1488	0.3525	3.2250	0.2115
Sz_H	2.8065	0.3376	0.9474	0.0621
Sz_A	2.7353	0.3418	0.9350	0.0613
Sz_M	2.6880	0.3381	0.9087	0.0596

Step 3: The standard deviation for each criterion is computed using the following formula:

$$\sigma_q = \sqrt{\frac{1}{n} \sum_{i=1}^n (x'_{iq} - \bar{x}'_q)^2} \quad (3)$$

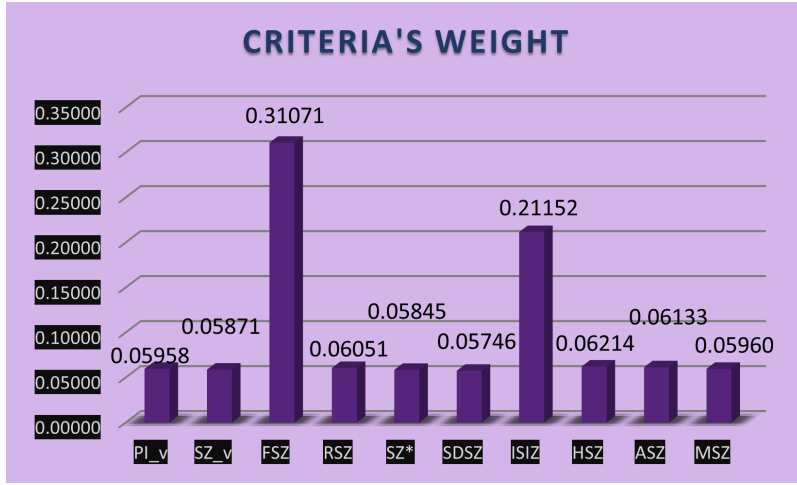


FIGURE 7 The relationship between reducible indices and their corresponding weights.

Where: - σ_q is the standard deviation of criterion q , - n is the number of alternatives, - $\bar{x}'_q$ is the mean of the normalized values for criterion q .

The correlation coefficient r_{ql} between criteria q and l is calculated using the following equation:

$$r_{ql} = \frac{\sum_{i=1}^n (x'_{iq} - \bar{x}'_q) (x'_{il} - \bar{x}'_l)}{\sqrt{\sum_{i=1}^n (x'_{iq} - \bar{x}'_q)^2 \sum_{i=1}^n (x'_{il} - \bar{x}'_l)^2}} \quad (4)$$

The information content C_q of criterion q is given by:

$$C_q = \sigma_q \cdot \left(1 - \frac{1}{m-1} \sum_{l=1}^m r_{ql} \right) \quad (5)$$

Where: - C_q is the information content of criterion q , - m is the number of criteria, - r_{ql} is the correlation between criteria q and l .

Finally, the weight w_q of each criterion q is calculated using the following equation. The resulting weights for each criterion are shown in Table 13:

$$w_q = \frac{C_q}{\sum_{q=1}^m C_q} \quad (6)$$

The weight w_q represents the relative importance of each criterion, and the sum of all weights equals 1, confirming the accuracy of our results. A graphical representation illustrating the relationship between the weight and the indices is shown in Figure 7.

6.2 | TOPSIS method

TOPSIS is a technique used to assess different options based on several criteria. It includes assigning weights to each criterion, normalizing the scores for comparison, and calculating the distance between each alternative and the ideal solution, which reflects the highest performance for each criterion [30], [31]. TOPSIS operates on the assumption that criteria should either consistently increase or decrease. Normalization is essential in multi-criteria scenarios to address the variations in units and scales among the criteria.

TABLE 14 Normalised decision matrix for TOPSIS method.

Drug Name	PI_v	Sz_v	Sz_F	Sz_R	Sz^*	Sz_{SD}	Sz_{ISI}	Sz_H	Sz_A	Sz_M
Carmustine	0.0213	0.0047	0.4315	0.0008	0.0037	0.0322	0.2903	0.0244	0.0010	0.0062
Afinitor	0.8582	0.9345	0.2788	0.9755	0.9499	0.8956	0.3353	0.8318	0.9792	0.9688
Etoposide	0.3403	0.2211	0.3174	0.1425	0.2282	0.3131	0.3854	0.3206	0.1130	0.1945
Temozolomide	0.0314	0.0090	0.4512	0.0018	0.0056	0.0374	0.3724	0.0430	0.0026	0.0063
Procarbazine	0.0413	0.0145	0.3384	0.0035	0.0089	0.0437	0.3167	0.0567	0.0050	0.0092
Tovorafenib	0.1703	0.0917	0.3361	0.0431	0.0804	0.1790	0.3539	0.1819	0.0423	0.0874
Belzutifan	0.1029	0.0409	0.3876	0.0147	0.0376	0.1211	0.3715	0.1075	0.0128	0.0524
Irinotecan	0.3240	0.2595	0.2326	0.1610	0.1941	0.2209	0.3905	0.3937	0.1626	0.1139
Weight	0.0596	0.0587	0.3107	0.0605	0.0584	0.0575	0.2115	0.0621	0.0613	0.0596

Step 1: To compute, we construct an evaluation matrix $(r_{iq})_{m \times k}$ that involves m alternatives and k criteria, where $i = 1, \dots, k$ and $q = 1, \dots, m$. This matrix enables us to represent the performance of each alternative across various criteria.

Step 2: Transform the evaluation matrix $(r_{iq})_{m \times k}$ into a normalized matrix $(Niq)_{m \times k}$ through the following formula.

$$\psi_{iq} = \frac{r_{iq}}{\sqrt{\sum_{i=1}^m x_{iq}^2}} \quad (7)$$

The values of this construction are presented in Table 14.

Step 3: Construct the weighted normalized matrix W , denoted as $(W_{iq})_{m \times k}$, by

$$W_{iq} = \psi_{iq} \times W_q \quad (8)$$

such as $w_1 + w_2 + w_3 + w_4 + w_5 + w_6 + w_7 + w_8$.

The values of this construction are presented in Table 15.

Step 4: Identify the optimal alternative E_b^+ and the non-optimal alternative E_w^- .

$$E_b^+ = \min\{W_{iq} | q \in V^-\}, \max\{W_{iq} | q \in V^+\} = W_{bq}^+ \quad (9)$$

$$E_w^- = \max\{W_{iq}|q \in V^-\}, \max\{W_{iq}|q \in V^+\} = W_{wq}^- \quad (10)$$

Here, $V^+ = q = 1, \dots, k|q$ identifies the optimal criteria, and $V^- = q = 1, \dots, k|q$ pertains to the worst criteria, illustrated in Table 15.

Step 5: Compute the squared Euclidean distances (L^2 distances) between each target alternative i and both the optimal alternative $(E_b)^+$ and the non-optimal alternative $(E_w)^-$.

$$P_{ib}^+ = \sum_{i=1}^n (W_{iq} - W_{bq}^+)^2 \quad P_{iw}^- = \sum_{i=1}^n (W_{iq} - W_{wq}^-)^2 \quad (11)$$

Step 6: Assess the similarity to the worst-case scenario for each respective target alternative.

$$Z_{iw} = \frac{P_{iw}^-}{P_{iw}^- + P_{ib}^+} \quad (12)$$

The value of variable Z_{iw} is set at 0 when the alternative signifies the optimal solution and at 1 it represents the worst solution.

Table 16 shows the required values using steps 4, 5 and 6.

Step 7: Arrange the alternatives in a ranked order based on the outcomes obtained from Table 16 .

TABLE 15 Weighted normalised decision matrix for TOPSIS method.

Drug Name	PI_v	Sz_v	Sz_F	Sz_R	Sz^*	Sz_{SD}	Sz_{ISI}	Sz_H	Sz_A	Sz_M
Carmustine	0.0013	0.0003	0.1341	0.0001	0.0002	0.0019	0.0614	0.0015	0.0001	0.0004
Afinitor	0.0511	0.0549	0.0866	0.0590	0.0555	0.0515	0.0709	0.0517	0.0600	0.0577
Etoposide	0.0203	0.0130	0.0986	0.0086	0.0133	0.0180	0.0815	0.0199	0.0069	0.0116
Temozolomide	0.0019	0.0005	0.1402	0.0001	0.0003	0.0021	0.0788	0.0027	0.0002	0.0004
Procarbazine	0.0025	0.0009	0.1051	0.0002	0.0005	0.0025	0.0670	0.0035	0.0003	0.0005
Tovorafenib	0.0101	0.0054	0.1044	0.0026	0.0047	0.0103	0.0749	0.0113	0.0026	0.0052
Belzutifan	0.0061	0.0024	0.1204	0.0009	0.0022	0.0070	0.0786	0.0067	0.0008	0.0031
Irinotecan	0.0193	0.0152	0.0723	0.0097	0.0113	0.0127	0.0826	0.0245	0.0100	0.0068

6.3 | VIKOR method

In the VIKOR methodology, we assess every available option against specified criteria to pinpoint the option that best approximates the ideal solution. After ranking the alternatives, we determine the compromise option that strikes the optimal balance and comes closest to the ideal solution. The notion of a compromise solution in multi-criteria decision making (MCDM) was first introduced by Zeleny and Yu[32], [33] in 1973 and its practical applications were demonstrated[34] in 1998.

Step 1: Compute the optimal best a^+ and the optimal worst a^- for each criterions, where q takes value from 1 to

TABLE 16 TOPSIS method with PA weighting:Computed Results.

Drug Name	P_{ib+}	P_{iw-}	Z_{iw}	TOPSIS rank
Carmustine	0.1561	0.0618	0.2836	4
Afinitor	0.0548	0.1555	0.7393	1
Etoposide	0.1258	0.0514	0.2898	3
Temozolomide	0.1537	0.0701	0.3132	2
Procarbazine	0.1575	0.0334	0.1751	8
Tovorafenib	0.1436	0.0393	0.2148	7
Belzutifan	0.1480	0.0520	0.2600	5
Irinotecan	0.1376	0.0447	0.2453	6

TABLE 17 Optimal best a^+ and the optimal worst a^- values for VIKOR method.

Drug Name	PI_v	Sz_v	Sz_F	Sz_R	Sz^*	Sz_{SD}	Sz_{ISI}	Sz_H	Sz_A	Sz_M
Carmustine	132	226	0.6467	2712	1132	72.13	1.8333	18.83	148.11	260
Afinitor	5321	45047	0.4179	3163260	287797	2005.31	2.1169	642.37	149629.19	40389
Etoposide	2110	10660	0.4757	462076	69146	700.97	2.4336	247.56	17273.39	8108
Temozolomide	195	433	0.6763	5720	1707	83.77	2.3513	33.17	397.26	261
Procarbazine	256	700	0.5072	11200	2696	97.85	2	43.75	762.23	384
Tovorafenib	1056	4421	0.5038	139854	24350	400.78	2.2348	140.49	6470.90	3642
Belzutifan	638	1972	0.5809	47508	11390	271.11	2.3461	83.00	1954.18	2184
Irinotecan	2009	12510	0.3487	521968	58803	494.52	2.4657	304.06	24840.12	4747
d^+	5321	45047	0.6763	3163260	287797	2005.31	2.4657	642.37	149629.19	40389
d^-	132	226	0.3487	2712	1132	72.13	1.8333	18.83	148.11	260

TABLE 18 Ranking of drugs using VIKOR method.

Drug Name	S_q	R_q	Q_q	VIKOR rank
Carmustine	0.717365	0.211523	0.718651	7
Afinitor	0.361742	0.245077	0.368465	3
Etoposide	0.559556	0.190257	0.490795	4
Temozolomide	0.512984	0.061224	0.177529	1
Procarbazine	0.787705	0.160381	0.698724	6
Tovorafenib	0.665633	0.163606	0.561897	5
Belzutifan	0.581299	0.090481	0.316354	2
Irinotecan	0.671363	0.310709	0.863436	8

k .

$$a^+ = \max\{a_{qI}, I = 1, \dots, L\}, \min\{a_{qI}, I = 1, \dots, L\} \quad (13)$$

for q^{th} beneficial function.

$$a^- = \min\{a_{qI}, I = 1, \dots, L\}, \max\{a_{qI}, I = 1, \dots, L\} \quad (14)$$

for q^{th} non-beneficial function. which is illustrated in Table 17.

Step 2: Calculate the weighted Chebyshev distance R_I and the weighted Manhattan distance S_q for each criterion, represented by $I = 1, \dots, L$.

$$S_q = \sum_{I=1}^m \left[w_q \times \frac{(a_q^+ - a_{qI})}{(a_q^+ - a_q^-)} \right] \quad R_q = \max \left[w_q \times \frac{(a_q^+ - a_{qI})}{(a_q^+ - a_q^-)} \right] \quad (15)$$

Step 3: Evaluate Q_I scores for each criterion, indicated by $I = 1, \dots, L$, employing the provided equation.

$$Q_I = \left[v \times \frac{(S_I - S^+)}{(S^- - S^+)} \right] + \left[(1 - v) \times \frac{(R_I - R^+)}{(R^- - R^+)} \right] \quad (16)$$

Step 4: Arrange the alternatives in their ranking orders based on the obtained results, as shown in Table 18

6.4 | COPRAS method

The COPRAS (COMplex PROportional ASsessment) method is a multi-criteria decision-making (MCDM) technique widely used for ranking and evaluating alternatives based on specific criteria. Below is a step-by-step explanation of the COPRAS method[39, 40].

Step 1: Define the Problem and Objectives. Clearly outline the problem requiring decision-making and identify the decision alternatives $A_1, A_2, \dots, A_m$. Define the criteria $C_1, C_2, \dots, C_n$, and specify whether each criterion is **beneficial** (higher values are better) or **non-beneficial** (lower values are better).

Step 2: Construct a decision matrix A where:

$$A = \begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}$$

Each element a_{ij} represents the performance of the i -th alternative under the j -th criterion.

Step 3: Normalize the decision matrix to make the criteria comparable:

- For **beneficial criteria**:

$$r_{ij} = \frac{a_{ij}}{\sum_{i=1}^m a_{ij}} \quad (17)$$

- For **non-beneficial criteria**:

$$r_{ij} = \frac{\min(a_{ij})}{a_{ij}} \quad (18)$$

The normalised decision matrix is found out in the Table 19.

Step 4: Assign weights w_j to each criterion, ensuring $\sum_{j=1}^n w_j = 1$. Calculate the weighted normalized values:

$$v_{ij} = r_{ij} \cdot w_j \quad (19)$$

And the weighted normalised matrix is presented in the Table 20.

Step 5: For each alternative A_i , calculate:

- The sum of **beneficial criteria**:

$$S_i^+ = \sum_{j \in B} v_{ij} \quad (20)$$

- The sum of **non-beneficial criteria**:

$$S_i^- = \sum_{j \in NB} v_{ij} \quad (21)$$

Here, B is the set of beneficial criteria indices, and NB is the set of non-beneficial criteria indices.

Step 6: Compute the relative utility U_i for each alternative:

$$U_i = \frac{S_i^+ + \frac{S_{\min}^-}{S_i^-} \cdot S_{\min}^-}{\sum_{i=1}^m \left(S_i^+ + \frac{S_{\min}^-}{S_i^-} \cdot S_{\min}^- \right)} \quad (22)$$

where $S_{\min}^-$ is the smallest value of S_i^- among all alternatives. Express U_i as a percentage.

Step 7: Rank the alternatives based on their utility degrees U_i . The alternative with the highest U_i value is the most favorable. (See Table 21)

Step 8: Perform a sensitivity analysis by varying the weights of the criteria or the decision matrix values to test the robustness of the results.

The graphical representation of brain tumor drugs evaluated using the TOPSIS method is shown in Figure 8. Similarly, the analysis using the VIKOR method is depicted in Figure 9, while the results derived from the COPRAS method are illustrated in Figure 10.

TABLE 19 Normalised decision matrix for COPRAS method.

Drug Name	PI_v	Sz_v	Sz_F	Sz_R	Sz^*	Sz_{SD}	Sz_{ISI}	Sz_H	Sz_A	Sz_M
Carmustine	0.0248	0.0050	0.9562	0.0009	0.0039	0.0360	0.7435	0.0293	0.0010	0.0064
Afnitor	1.0000	1.0000	0.6179	1.0000	1.0000	1.0000	0.8585	1.0000	1.0000	1.0000
Etoposide	0.3965	0.2366	0.7034	0.1461	0.2403	0.3496	0.9870	0.3854	0.1154	0.2007
Temozolomide	0.0366	0.0096	1.0000	0.0018	0.0059	0.0418	0.9536	0.0516	0.0027	0.0065
Procarbazine	0.0481	0.0155	0.7500	0.0035	0.0094	0.0488	0.8111	0.0681	0.0051	0.0095
Tovorafenib	0.1985	0.0981	0.7449	0.0442	0.0846	0.1999	0.9064	0.2187	0.0432	0.0902
Belzutifan	0.1199	0.0438	0.8589	0.0150	0.0396	0.1352	0.9515	0.1292	0.0131	0.0541
Irinotecan	0.3776	0.2777	0.5156	0.1650	0.2043	0.2466	1.0000	0.4733	0.1660	0.1175

TABLE 20 Weighted normalised decision matrix for COPRAS method.

Drug Name	PI_v	Sz_v	Sz_F	Sz_R	Sz^*	Sz_{SD}	Sz_{ISI}	Sz_H	Sz_A	Sz_M
Carmustine	0.0015	0.0003	0.2971	0.0001	0.0002	0.0021	0.1573	0.0018	0.0001	0.0004
Afnitor	0.0596	0.0587	0.1920	0.0605	0.0584	0.0575	0.1816	0.0621	0.0613	0.0596
Etoposide	0.0236	0.0139	0.2185	0.0088	0.0140	0.0201	0.2088	0.0239	0.0071	0.0120
Temozolomide	0.0022	0.0006	0.3107	0.0001	0.0003	0.0024	0.2017	0.0032	0.0002	0.0004
Procarbazine	0.0029	0.0009	0.2330	0.0002	0.0005	0.0028	0.1716	0.0042	0.0003	0.0006
Tovorafenib	0.0118	0.0058	0.2315	0.0027	0.0049	0.0115	0.1917	0.0136	0.0027	0.0054
Belzutifan	0.0071	0.0026	0.2669	0.0009	0.0023	0.0078	0.2013	0.0080	0.0008	0.0032
Irinotecan	0.0225	0.0163	0.1602	0.0100	0.0119	0.0142	0.2115	0.0294	0.0102	0.0070

TABLE 21 Ranking of drugs using COPRAS method.

Drug Name	S_i	U_i	COPRAS rank
Carmustine	0.4608	0.4608	7
Afnitor	0.8514	0.8514	1
Etoposide	0.5508	0.5508	2
Temozolomide	0.5218	0.5218	3
Procarbazine	0.4170	0.4170	8
Tovorafenib	0.4815	0.4815	6
Belzutifan	0.5009	0.5009	4
Irinotecan	0.4932	0.4932	5

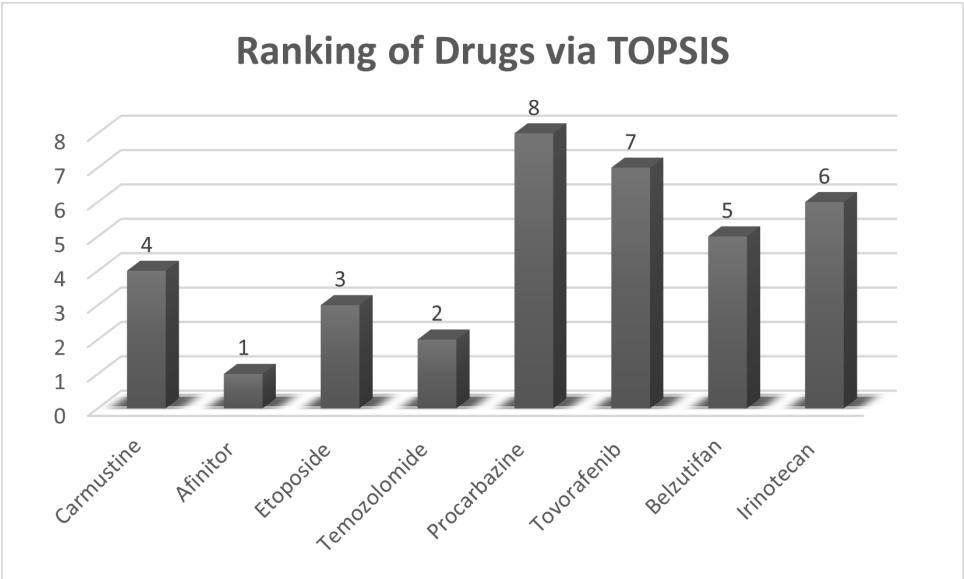


FIGURE 8 Ranking of Drugs via TOPSIS

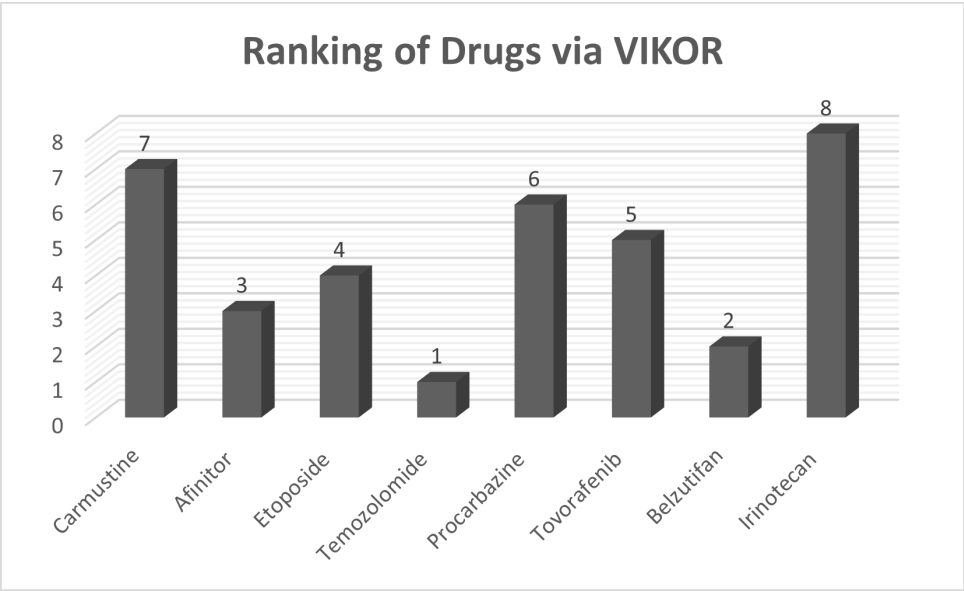


FIGURE 9 Ranking of Drugs via VIKOR

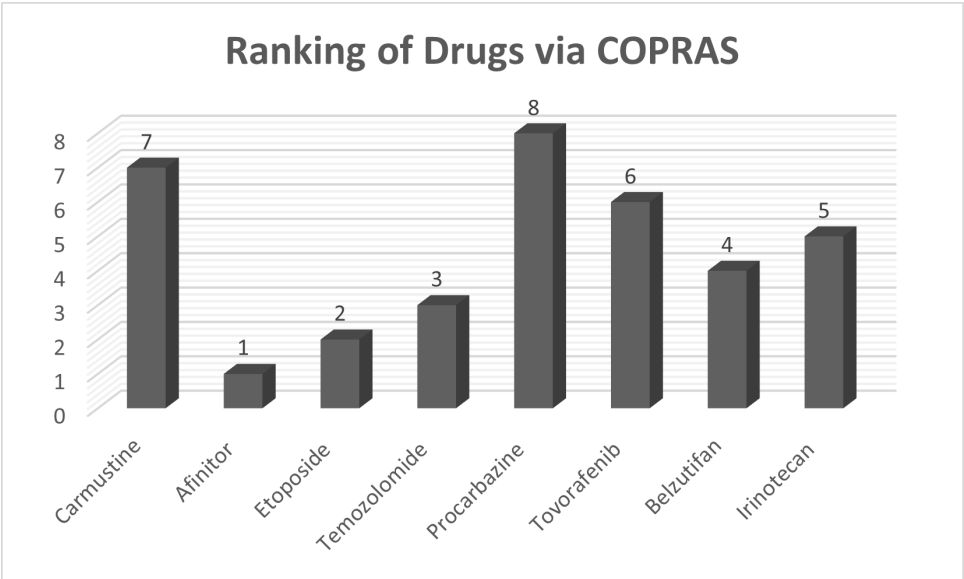


FIGURE 10 Ranking of Drugs via COPRAS

6.5 | Assessment between TOPSIS, VIKOR and COPRAS

A comparison of the results obtained from the TOPSIS, VIKOR and COPRAS ranking methods for the eight drugs revealed notable differences in the prioritization of the medications, as illustrated in the Table 22. Afinitor emerged

TABLE 22 Assessment between TOPSIS, VIKOR and COPRAS method.

Method	Ranking	Best	Worst
TOPSIS	Afinitor>Temozolomide>Etoposide>Carmustine>Belzutifan>Irinotecan>Tovorafenib>Procarbazine	Afinitor	Procarbazine
VIKOR	Temozolomide>Belzutifan>Afinitor>Etoposide>Tovorafenib>Procarbazine>Carmustine>Irinotecan	Temozolomide	Irinotecan
COPRAS	Afinitor>Etoposide>Temozolomide>Belzutifan>Irinotecan>Tovorafenib>Carmustine>Procarbazine	Afinitor	Procarbazine

as the top-performing medication in the TOPSIS analysis, closely followed by Temozolomide and Etoposide. In the VIKOR method, Temozolomide ranked highest, with Belzutifan and Afinitor securing the next positions. The primary goal of the VIKOR technique was to identify a compromise solution that satisfied all stakeholders while minimizing dissatisfaction. This analysis highlighted Temozolomide, Belzutifan, and Afinitor as the most favorable options, as they consistently performed well across various evaluated criteria.

Using the COPRAS approach, Afinitor was also identified as the leading choice, followed by Etoposide and Temozolomide. COPRAS prioritized medications that showed strong overall performance, even if they did not excel in every individual category. This method provided insights into identifying the most reliable and effective drugs overall.

All three techniques—TOPSIS, VIKOR, and COPRAS—produced consistent results regarding Afinitor’s ranking. It was rated first in both TOPSIS and COPRAS and placed third in VIKOR, reflecting slight differences in evaluation criteria but overall agreement on its efficacy. Meanwhile, Procarbazine was ranked eighth by both COPRAS and TOPSIS. Figures 8, 9, and 10 illustrate the graphical representation of the rankings derived from the three methodologies.

7 | CONCLUSION

The analysis of polynomial QSPR models and topological indices reveals notable relationships between specific drug properties and various TIs. With a correlation coefficient of ($r = 0.9866$), the *PI* index demonstrates a strong association with molar volume. Similarly, the *ISIZ* index exhibits a high correlation ($r = 0.9888$) with polarizability and also shows significant relevance to molar refractivity. Additionally, the *ISIZ* index emerges as a key predictor, attaining the highest correlation coefficient ($r = 0.9874$) for molar volume.

By applying the MCDM method and integrating the CRITIC technique with TOPSIS, VIKOR, and COPRAS, we have obtained valuable insights into the comparative assessment of eight drugs used in the treatment of brain tumors. This approach has garnered significant attention from researchers in recent years. TOPSIS and VIKOR provide effective strategies for addressing MCDM problems, often leading to decisions that closely align with the ideal solution and are widely accepted by decision-makers. The CRITIC method is employed to determine weights, aiding decision-makers in evaluating the significance of drug interventions for various applications.

Our final analysis identified Afinitor as the highest-ranked drug according to both the TOPSIS and COPRAS methods, making it the most suitable choice for approximating the ideal solution. Meanwhile, the VIKOR method ranked Temozolomide as the top drug. The theoretical advancements in ranking drug structures based on chemical invariants hold great significance for biomedicine and mathematical chemistry, contributing to the enhancement of drug discovery and development processes.

DATA AVAILABILITY

In this article, no data were utilized.

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CONFLICTS OF INTEREST

Authors have no conflict of interest

AUTHORS' CONTRIBUTIONS

All authors equally contributed to this work. All authors read and approved the final manuscript.

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